Monopolistically Skewed Business Cycles*

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Abstract

How do aggregate shocks reshape the distribution of firm growth? We show that mean growth, dispersion, and skewness can be linked to two structural objects: firms' heterogeneous exposure to aggregate shocks and the local curvature of output policies. Heterogeneous exposures determine cross-sectional dispersion after a shock; curvature determines whether the distribution is symmetric or skewed. The size gradient in skewness points to the source of this curvature: large firms exhibit larger skewness fluctuations, consistent with market power. We microfound this channel using demand-side primitives and show that moving from price taking toward market power makes the equilibrium local curvature index more negative. In quarterly U.S. public-firm data, skewness cyclicality rises across estimated-markup and profitability deciles and with industry concentration; firms with greater pandemic exposure display larger skewness responses; and six identified shocks generate tight growth–skewness comovement. Market structure turns aggregate shocks into asymmetric business cycles.

Keywords: business cycle, skewness, market power

JEL Codes: D21, E32, L11

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1 Introduction

Are higher moments of firm growth primitive features of business cycle shocks, or do they emerge from firms' propagation of aggregate disturbances? This paper argues for the latter. A single aggregate shock can move not only average firm growth, but also dispersion and skewness across firms, once two empirically natural ingredients are present: firms differ in their exposure to aggregate shocks, and product-market competition makes firms' output policies locally concave in those shocks.

We start from a set of distributional facts about firm growth. Using quarterly data on U.S. public firms, we show that the cross-sectional distribution of real sales growth changes shape systematically over the business cycle. Mean growth falls sharply in recessions, with no comparable acceleration in expansions.[1] Dispersion rises when aggregate growth moves away from trend, increasing in both large contractions and large expansions. Skewness is procyclical: the distribution becomes left-skewed in downturns and right-skewed when growth is strong.[2] The most striking feature is that these skewness movements are not uniform across firms. Their cyclical amplitude is more than twice as large among large firms as among small firms. We call this pattern the *size gradient of skewness.*

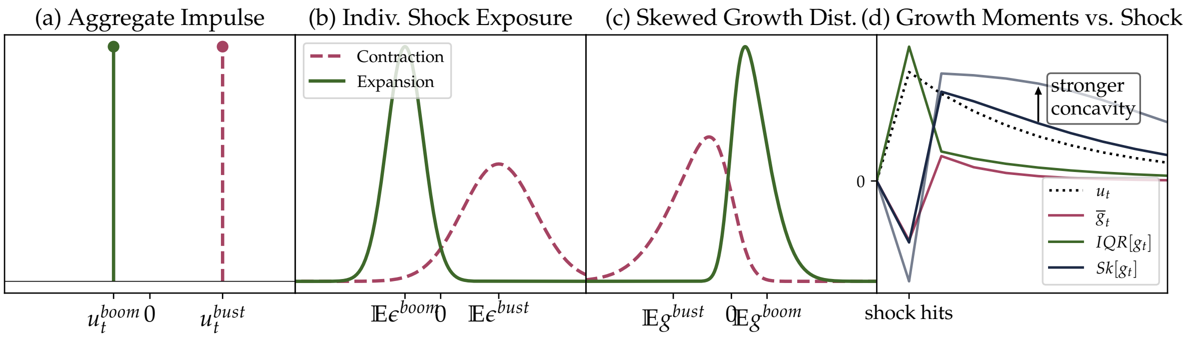
These facts pose a challenge for business-cycle modeling. A natural way to fit movements in different moments of the firm-growth distribution is to introduce different shock processes: first-moment shocks for average growth, uncertainty shocks for dispersion, and skewness or tail-risk shocks for asymmetry. Such an approach can be useful, but it leaves open a deeper question. Do higher moments move because shocks themselves have time-varying dispersion and skewness, or because firms' responses transform ordinary aggregate shocks into fluctuating higher moments of growth? We take the second view and develop a propagation mechanism through which a mean-shifting aggregate shock generates movements in the first, second, and third moments of firm growth jointly.

The mechanism is simple. Suppose firms are hit by a common aggregate disturbance but differ in their exposure to it (e.g. Davis et al., 2025). The common component moves firms in the same direction and shifts average growth. Differences in exposure determine how the aggregate shock is distributed across firms. When the aggregate factor changes sharply, whether in a contraction or a rebound, high-exposure firms receive larger effective shocks than firms closer to the center of the exposure distribution. Heterogeneous exposure therefore turns aggregate shocks into cross-sectional dispersion of idiosyncratic shocks. The second ingredient is curvature. If firms' log-output policies are locally concave in shocks, the same cross-sectional distribution of shock incidence maps into an asymmetric distribution of growth rates. On impact of an adverse aggregate shock, ad-

[1] This asymmetry has been documented at least since Sichel (1993).

[2] Salgado et al. (2025) first emphasized this procyclicality. We use the term left-skewed (right-skewed) synonymously with negatively skewed (positively skewed).

Figure 1: Distributional Shifts in Theory



Note: The figure shows how an aggregate impulse, u_t, maps into heterogeneous firm shocks in (a)–(b), then into a skewed growth-rate distribution through log-concavity in (c). Panel (d) plots the implied co-movement of mean growth $\overline{g}$, IQR, and Kelley skewness after an AR(1) aggregate shock.

versely exposed firms contract sharply and form the long left tail. Firms that are weakly or favorably exposed do not generate an equally long right tail, because concavity compresses positive responses. The distribution therefore becomes negatively skewed. As the shock unwinds, the firms hit hardest on impact exhibit the largest subsequent recoveries, creating a right tail and turning skewness positive. Figure 1 illustrates the logic: the aggregate impulse moves the mean, heterogeneous exposure generates dispersion, and local concavity maps that dispersed incidence into skewness. Panel (d) then shows the implied path after a mean-reverting shock, with mean growth, dispersion, and skewness moving together over the cycle.

We summarize this asymmetry by a local curvature index: the normalized second derivative of the firm's log-output policy. Negative values correspond to local concavity, and more negative values mean that the same dispersed shock incidence is mapped into a more asymmetric growth distribution. Holding the shock process fixed, this index therefore scales the amplitude of the skewness response. The size gradient therefore becomes a restriction on primitives: holding the shock process fixed, larger skewness swings among large firms require policies that are more negatively curved, or equivalently, more locally concave.

We argue that product-market power provides exactly this primitive. In a price-setting problem with isoelastic costs and variable-elasticity demand, the local curvature of a firm's log-output policy is governed by the shape of marginal revenue. Price-taking firms have no markup-adjustment margin, so their policies are locally linear. As a result, a cross section of price takers facing symmetric shock incidence produces a symmetric distribution of growth rates. Price-setting firms behave differently. Favorable shocks are absorbed partly through prices and markups, while adverse shocks translate more strongly into quantity declines. We provide primitive conditions under which this asymmetry makes log-output policies locally concave: a strengthened version of Marshall's Second Law, together with increasing pass-through, is sufficient. These are familiar demand restrictions rather than a new shock process: they discipline how elasticities and markups change along the demand curve.

We then isolate the role of market power by moving continuously from price taking to price setting. At the competitive benchmark, curvature is zero and the second-order skewness channel is absent. As market power is introduced, the same demand-side conditions make equilibrium curvature more negative. Holding shocks fixed, more negative curvature then amplifies cyclical movements in skewness. CES demand is the knife-edge case in which the channel remains shut down, because constant elasticity eliminates the relevant local curvature. Away from CES, variable-elasticity demand naturally generates the curvature needed for aggregate shocks to produce asymmetric growth distributions.

This mechanism provides a natural interpretation of the size gradient of skewness. Large firms are more likely to operate with market power and to face variable-elasticity demand.[3] If market power makes log-output policies more concave, then large firms should display larger cyclical swings in skewness even when hit by the same aggregate shock. Size is not the primitive force in the theory; it is a proxy for the curvature generated by market power. We therefore complement the size split with estimated markups and with profitability and industry concentration as additional measures related to market power.

Our framework delivers three empirical predictions, which we test using quarterly Compustat data from 1983 to 2024. First, skewness fluctuations should be stronger in environments where market power is greater. Second, firms with greater exposure to an aggregate shock should exhibit larger skewness responses. Third, aggregate shocks should generate tight comovement between mean growth and skewness.

We begin by testing the market-power interpretation using estimated markups, profitability, and industry concentration. Using production-based markup estimates following De Loecker et al. (2020), we find that the skewness–growth coefficient rises significantly across markup deciles. The gradient is even stronger across profitability deciles, while more concentrated industries exhibit larger fluctuations in skewness. Because production-based markups constructed from revenue data are measured with error and may incorporate demand-induced price variation, we view the three measures as complementary. Together, they support the interpretation that the size gradient of skewness reflects market power rather than size alone.

We next use heterogeneity in firms' pre-pandemic risk-factor disclosures as a source of differential exposure to the COVID shock. Firms with greater pandemic exposure experience a larger decline in skewness at the onset of the pandemic, followed by a positive rebound during the recovery. This impact-and-recovery pattern mirrors the model's prediction: the same concavity that produces negative skewness when the shock hits

[3] The connection between size and market power has also been shown to hold empirically; see, for example, Autor et al. (2020).

produces positive skewness as the shock unwinds.

Finally, we estimate local projections for six identified aggregate shocks: monetary, oil, credit, uncertainty, sentiment, and TFP shocks. Across all six shocks, contractionary disturbances lower cross-sectional skewness of firm sales growth, and the skewness response closely tracks the response of aggregate growth. The correlation between the impulse responses of growth and skewness ranges from 0.89 to 0.98, showing that the comovement is not specific to one recession episode, identification strategy, or type of disturbance. Splitting these responses by firm size reveals that the skewness response is concentrated among large firms: their skewness falls substantially after contractionary shocks, while the response among small firms is close to zero. This pattern is exactly what the curvature mechanism predicts if large firms operate with greater market power.

The paper contributes to the literature in three ways. First, it provides a parsimonious, shock-agnostic explanation of mean, dispersion, and skewness dynamics in firm growth. Existing work has documented asymmetric aggregate cycles, countercyclical firm-level dispersion, procyclical skewness, and size-dependent cyclicality. We show that these patterns need not be assigned to separate first-, second-, and third-moment shocks: a single aggregate disturbance, combined with heterogeneous exposure and demand-side curvature, can move all three moments together.

Second, the paper microfounds the curvature that generates skewness dynamics in growth rates. Ilut et al. (2018) show that concave hiring rules can transform shocks into skewed employment growth, treating the asymmetric response as a reduced-form mapping from shocks to growth. We ask where such curvature comes from in firm sales and how it operates when growth rates are first differences. We derive a concave log-output policy from firms' price-setting problem; sales growth then compares current and lagged positions on this policy. Skewness therefore depends not only on the policy's curvature, but also on how current shock incidence differs from past shock incidence. The curvature itself is governed by familiar demand-system objects: elasticities, superelasticities, pass-through, and markups. These objects are usually studied in industrial organization and trade, but here they determine higher moments of firm growth over the business cycle.

Third, the paper highlights a dynamic macroeconomic role for market structure. Market power does not only affect static markups, allocative efficiency, or reallocation; it also shapes how aggregate shocks are distributed across firms and how downturns are amplified through the left tail. Because the mechanism is grounded in firm-level demand curvature, the size gradient disciplines alternative explanations: supply-side, network, or other aggregate mechanisms may also generate skewness, but they must explain why the relevant curvature is systematically stronger for large firms and for firms with greater market-power proxies. In our framework, market structure turns aggregate shocks into distributional shocks.

Literature Our paper is closest to Salgado et al. (2025), who document procyclical skewness in firm outcomes, and Ilut et al. (2018), who show that concave firm policies can turn symmetric shocks into skewed growth responses. We share the propagation view, but ask where the relevant concavity comes from and why it should be stronger for large firms. Related work treats skewness as a macroeconomic state variable or forecasting object (Iseringhausen et al., 2026; Dew-Becker, 2024), or studies other sources of aggregate asymmetry (Kamepalli et al., 2025; Dew-Becker and Vedolin, 2021). Our mechanism instead operates at the firm level: heterogeneous exposure to aggregate shocks generates dispersion, while demand-side curvature maps that dispersion into realized sales-growth skewness.

Firm heterogeneity and exposure. The empirical facts build on work showing that business cycles and firm growth are distributional. Business cycles are asymmetric (Sichel, 1993; McKay and Reis, 2008; Morley and Piger, 2012; Berger et al., 2020; Dupraz et al., 2025), firm-growth distributions have higher moments that move over the cycle (Higson et al., 2002, 2004), and the firm-growth literature around Gibrat's Law shows that growth is not simply scale invariant or Gaussian (Sutton, 1997; Bottazzi and Secchi, 2006; Sterk et al., 2021). Firm responses also vary systematically by size and age (Gertler and Gilchrist, 1994; Fort et al., 2013; Crouzet and Mehrotra, 2020). Our empirical implementation is closest to Davis et al. (2025), who measure heterogeneous firm exposure to macro shocks. We use such exposure heterogeneity to connect aggregate shocks to dispersion, complementing uncertainty and endogenous-dispersion accounts of cross-sectional fluctuations (Bloom, 2009; Arellano et al., 2019; Schaal, 2017; Bloom et al., 2018; Kehrig, 2015; Decker et al., 2016; Straub and Ulbricht, 2019).

Demand theory and market power. The microfoundation draws on non-CES demand, variable markups, and pass-through. Marshall's Second Law, superelasticities, and incomplete pass-through determine how shocks map into quantities under market power (Feenstra, 2003; Parenti et al., 2017; Mrázová and Neary, 2017; Melitz, 2018; Matsuyama and Ushchev, 2022; Weyl and Fabinger, 2013), with empirical support from product-mix, exchange-rate pass-through, and strategic-complementarity evidence (Berman et al., 2012; Amiti et al., 2019; Mayer et al., 2021). Since the channel links curvature to market power, it also connects to work on firm size, markups, concentration, and reallocation toward high-markup firms (Hall, 1986; De Loecker and Warzynski, 2012; Autor et al., 2020; Baqaee et al., 2024; Burstein et al., 2025). Our contribution is a complementary business-cycle implication: market power shapes not only average markups, pass-through, or reallocation, but also the skewness of firm growth over the cycle.

Outline Section 2 documents the stylized facts. Section 3 shows how heterogeneous exposures and concave policies map aggregate shocks into higher moments of firm growth. Section 4 microfounds this curvature using imperfect competition and variable-elasticity demand. Section 5 tests the predictions in the data. Section 6 concludes.

2 Stylized Facts, Stylized Mechanisms

Before turning to the formal theory, we introduce the reduced-form policy representation that guides the paper. Let q_{it} denote real sales of firm i in quarter t, and let g_{it} denote real sales growth. We study the cross-sectional distribution of g_{it}, summarized by its mean $\bar{g}_t$, interquartile range $s_t = g_{t,0.75} - g_{t,0.25}$, and Kelley skewness

$$\mathrm{Sk}[g_t] := \frac{g_{t,0.9} + g_{t,0.1} - 2g_{t,0.5}}{g_{t,0.9} - g_{t,0.1}}, \tag{1}$$

where $g_{t,p}$ is the p-th quantile of the time-t cross section. Negative values indicate a longer left tail; positive values indicate a longer right tail. The representation separates common movements in output from changes in the shape of the cross section.

Consider a cross section of firms. Firm i's log output is governed by a policy $Q^\star$ and two shock components: a centered firm-level component ϵ_{it}, with $\mathbb{E}[\epsilon_{it}] = 0$, and an aggregate component ξ_t common to all firms:

$$\log q_{it} = Q^\star(\epsilon_{it} + \xi_t \mid \epsilon_{-i,t} + \xi_t) + at = Q^\star(\epsilon_{it}) - b\,\xi_t + at. \tag{2}$$

Here $a \geq 0$ is a secular trend. The second equality imposes homotheticity: the aggregate shock changes the common scale of output, while the shape of the cross section is governed by the distribution of ϵ_{it} and by the curvature of $Q^\star$.[4] Since we speak of cost shocks, higher ξ_t lowers output, so $b > 0$. Taking first differences gives

$$g_{it} = a + \Delta Q^\star(\epsilon_{it}) - b\,\Delta\xi_t. \tag{3}$$

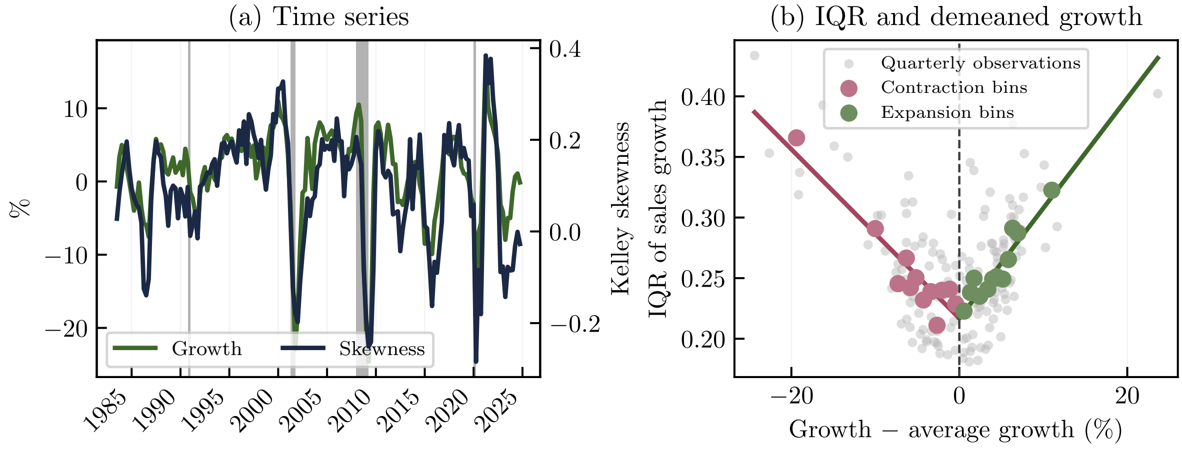
Equation (3) preserves this separation in growth rates: $-b\,\Delta\xi_t$ shifts the center of the growth distribution, while $\Delta Q^\star(\epsilon_{it})$ determines its shape through shock incidence and policy curvature. We use this decomposition to organize the first three moments of firm growth in our 1983-2024 Compustat sample. We take q_{it} to be quarterly real sales. Nominal sales are deflated by the GDP deflator, and growth is measured as year-over-year log growth.[5]

Mean growth In the language of (3), recessions are episodes in which the aggregate component ξ_t rises sharply. Since this term is common across firms, it shifts the growth distribution left and lowers average growth. Panel (a) of Figure 2 shows this pattern

[4] For a discussion of monopolistic competition with homothetic demand, see Parenti et al. (2017). Supplemental Appendix B provides a homothetic monopolistic-competition environment that delivers the decomposition in (2).

[5] Appendix C describes the sample construction and cleaning rules in detail.

Figure 2: Growth, Skewness, and Dispersion Around Trend



Note: Panel (a) plots aggregate real sales growth and Kelley skewness of the cross-sectional distribution of firm sales growth. Panel (b) plots the interquartile range of firm sales growth against aggregate growth relative to its sample mean. Gray dots are quarterly observations; colored dots are binned averages for contractions and expansions.

directly. Aggregate sales growth is usually close to trend, but occasionally falls sharply. The sharp drops are not mirrored by equally large cyclical movements during expansions, a form of business-cycle asymmetry documented by Sichel (1993) and Morley and Piger (2012). In particular, Morley and Piger (2012) find that cyclical variation in U.S. output is substantially larger during recessions than during expansions and reject linearity in favor of nonlinear specifications capturing asymmetric recessions and recoveries.

Stylized Fact 1

Mean growth rates usually fluctuate around trend but occasionally drop sharply at the onset of recessions.

Dispersion Equation (3) imposes a simple restriction: the common aggregate term shifts all firms together and therefore cannot by itself widen the growth distribution. Dispersion must come from the firm-specific term, $\Delta Q^{\star}(\epsilon_{it})$. One possible explanation is a location channel: aggregate conditions move firms to a steeper part of the policy function, so the same cross-sectional shock incidence generates more dispersed growth.[6] Taken on its own, this channel predicts a monotone relationship between dispersion and the aggregate state. If high-cost recession states place firms on steeper parts of the policy, dispersion should rise in contractions and fall in expansions.

Panel (b) of Figure 2 does not show such a purely monotone relationship. The horizontal axis is aggregate growth relative to its sample mean, and the vertical axis is the interquartile range of firm growth. Dispersion is lowest near average growth and rises on both sides. Appendix Table A.1 confirms the same pattern: the unconditional rela-

[6]This is the approach taken by Ilut et al. (2018).

tionship between dispersion and growth is weak, while the split-slope specification gives a negative IQR-growth slope below average growth and a positive slope above average growth.

This V-shape points to changing shock incidence rather than a pure location channel. In the theory below, aggregate shocks generate dispersion because firms have heterogeneous exposure to the same aggregate factor. What matters for growth dispersion is the change in that factor, not only its level. A sharp contraction and a sharp rebound both spread firms out according to their exposures, so dispersion rises when aggregate growth moves far from trend in either direction.

Stylized Fact 2

Growth dispersion follows the absolute deviation of aggregate growth from trend: it is low near trend and rises in contractions and expansions alike.

Skewness As emphasized by Salgado et al. (2025), firm-growth skewness is strongly procyclical. Panel (a) of Figure 2 shows the same pattern in our data: firm-growth distributions become more left-skewed in downturns and more right-skewed when aggregate growth is strong.[7]

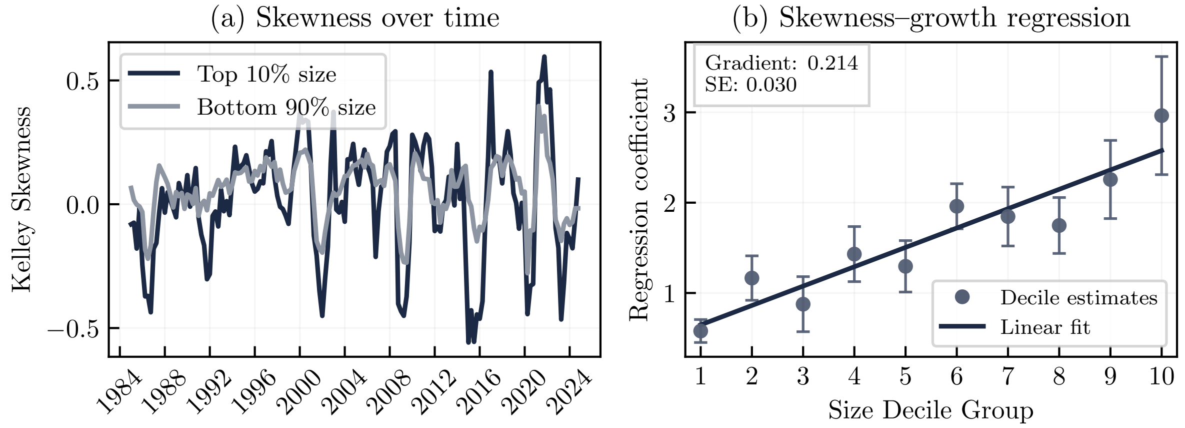
We interpret this pattern through the curvature of $Q^\star$. If $Q^\star$ were affine, a wider but symmetric cross section of ϵ_{it} would raise dispersion without systematically changing skewness. Concavity of $Q^\star$ explains why dispersion can also become asymmetry. Consider the simple case in which the economy is near trend in $t-1$ and firms receive a dispersed adverse cost shock in t. Then, up to the common shift,

$$g_{it} = Q^\star(\epsilon_{it}) - Q^\star(0) - b\,\xi_t.$$

The common term shifts all firms together and does not affect skewness. The shape of the growth distribution is therefore governed by $Q^\star(\epsilon_{it})$. If $Q^\star$ is decreasing and locally concave, adverse firm-level shocks generate large log-output declines, while favorable shocks are compressed into smaller gains. A symmetric widening of ϵ_{it} is then mapped into a left-skewed distribution of growth rates when the aggregate shock hits. The key driver of growth-rate skewness is shock dispersion. In the example above, ϵ_{it} is dispersed while the lagged cross section is close to zero, so the sign and magnitude of skewness in $Q^\star(\epsilon_{it})$ determine the skewness of g_{it}. If the shock unwinds in the next period, growth rates approximately reverse and skewness turns positive. More generally, the period with larger cross-sectional dispersion in ϵ dominates the sign and magnitude of growth

[7]We do not interpret the average level of skewness as the central object. A small positive long-run level may reflect secular innovation, entry and exit, or skewed idiosyncratic shocks. For innovation-based sources of right-skewed growth, see, for example, the literature on quality-ladder models or, recently, Berlingieri et al. (2025). The relevant fact for our mechanism is the cyclical movement of skewness with aggregate growth.

Figure 3: Size-dependent skewness



Note: Size groups are defined using average real sales over the previous three years. The standard deviation of Kelley skewness is about 0.23 for large firms, compared with 0.12 for small firms. In Panel (b), points are decile-specific skewness-growth coefficients, bars are 95% confidence intervals, and the line is the fitted linear gradient across deciles. We use Newey–West standard errors with four quarterly lags.

skewness. Combined with Stylized Fact 2, which says that dispersion rises when growth departs from trend, this yields negative skewness in contractions and positive skewness in recoveries.

This interpretation differs from location-shift mechanisms such as Ilut et al. (2018). In those models, higher moments move because the economy evaluates a nonlinear growth-response rule at different aggregate states. Under our homothetic representation, common location shifts are absorbed by ξ_t and are neutral for skewness. Cyclical skewness instead comes from a fixed log-output policy that maps current and past shock incidence into asymmetric growth outcomes through first differences.

The size gradient The most restrictive fact is not just that skewness is procyclical, but that the amplitude of this procyclicality differs strongly across firms. Figure 3 shows this in two ways. Panel (a) splits firms by size and plots Kelley skewness separately for large firms and for the rest of the Compustat sample. Both series move with the cycle, but the large-firm series has much larger swings. Panel (b) assigns firms to size deciles in each quarter. The points report skewness-growth coefficients estimated separately for each decile. To summarize their relationship with size, the fitted line comes from a pooled specification in which the skewness-growth coefficient varies linearly with the decile rank. Its estimated gradient is 0.214 per decile and is highly statistically significant. Thus, the pattern across all ten deciles shows that the result is not an artifact of the top-ten-percent split. Appendix Table A.1 reports the same size gradient in the baseline regressions: the skewness-growth slopes are substantially larger for large firms than for small firms.

This gradient is a restriction on $Q^\star$. If large and small firms faced the same aggregate timing and differed only in the common term $-b\,\Delta\xi_t$, their skewness would not move with different amplitudes. In the framework above, larger skewness amplitude points to stronger curvature of the policy mapping. Size is not the primitive economic force in our interpretation; it is a first proxy for market power.[8]

Stylized Fact 3

Skewness is strongly procyclical, and its cyclical amplitude is substantially larger for large firms.

The preceding facts point to the reduced form we carry into the theory. We treat growth as the first difference of a policy for log output, rather than as a static nonlinear map from a time-t signal into growth.[9] This formulation keeps two margins separate: the common aggregate components ξ_t and ξ_{t-1}, which move the scale of output, and the centered cross-sectional incidences ϵ_{it} and ϵ_{it-1}, which change the shape of the growth distribution.

This separation makes clear what the theory must explain. Heterogeneous exposure determines how aggregate shocks spread firms out across the policy, while curvature determines whether those changes in current and lagged shock incidence become asymmetric growth outcomes. The next section formalizes this reduced form and shows how it links aggregate shocks to mean growth, dispersion, and skewness. Section 4 then derives the relevant curvature, and its market-power gradient, from demand-side primitives.

3 Firm Policies and the Growth Distribution

We now formalize the reduced form introduced above. A shock process, together with the local shape of the log-output policy $Q^\star$, determines the cross-sectional moments of firm growth. We keep $Q^\star$ as a reduced-form object, derive approximate expressions for mean growth, dispersion, and skewness, and isolate the local curvature index that the demand-side microfoundation must explain.

Exposure structure. Section 2 separated growth into a common component and a centered component. We now give these two components a factor structure. There is a continuum of firms, $i \in I$, and an aggregate factor u_t. Firm i's total exposure to the aggregate factor is $\widetilde{\lambda}_i u_t$, where $\widetilde{\lambda}_i$ is a firm-specific loading. The average loading,

[8] The connection between size and market power has also been shown to hold empirically; see, for example, Autor et al. (2020).

[9] This differs from Ilut et al. (2018) and Straub and Ulbricht (2019), who write growth as $g_{it} = f(s_{it})$ for a time-t signal s_{it} and a concave map f. In that class of models, higher moments move because the economy shifts the location at which a nonlinear growth-response rule is evaluated. Our formulation instead keeps a fixed log-output policy and lets growth compare current and lagged positions on that policy.

$\overline{\lambda} = \mathbb{E}[\widetilde{\lambda}] \geq 0$, captures the part of the aggregate factor that moves all firms together. Deviations from that average capture the part that spreads firms out around the cross-sectional mean. We therefore define centered exposure by

$$\lambda_i := \widetilde{\lambda}_i - \overline{\lambda},$$

with symmetric density Λ and normalization $\mathbb{V}(\lambda) = \int \lambda^2 \Lambda(\lambda)\, d\lambda = 1$. In addition to its aggregate exposure, firm i receives a purely idiosyncratic shock $\kappa_{i,t}$, which is independent across firms and time, has mean zero, and has variance $\sigma^2_{\kappa,t}$.

The resulting shock to firm i's marginal cost is

$$\widetilde{\epsilon}_{i,t} = \kappa_{i,t} + \underbrace{\lambda_i u_t}_{\text{centered aggregate incidence}} + \underbrace{\overline{\lambda} u_t}_{\text{common component}}, \qquad \overline{\lambda}, u_t \geq 0. \tag{4}$$

It scales marginal cost by $\exp(\widetilde{\epsilon}_{i,t})$.[10] The common component, $\overline{\lambda} u_t$, is the aggregate shock ξ_t from Section 2; it shifts all firms in the same direction. The centered component, $\kappa_{i,t} + \lambda_i u_t$, determines where firm i sits relative to the cross-sectional mean. The level of u_t is nonnegative, so a higher u_t is contractionary on average, but growth depends on changes in u_t: u_t rises when the shock hits and falls during the recovery.

Let $Q^\star$ denote the firm's log-output policy as in Section 2. Substituting the factor structure into (2) and taking first differences yields growth rates

$$g_{i,t} = a + \Delta Q^\star(\kappa_{i,t} + \lambda_i u_t) - b\,\overline{\lambda}\,\Delta u_t. \tag{5}$$

This expression makes the two margins transparent. The common term $-b\,\overline{\lambda}\,\Delta u_t$ moves the center of the growth distribution. The term $\Delta Q^\star(\kappa_{i,t} + \lambda_i u_t)$ governs its shape. When u_t changes sharply, firms with different λ_i's move by different amounts across the policy. Whether this cross-sectional spread remains symmetric or becomes skewed depends on the local curvature of $Q^\star$.

Moment approximation. The next proposition formalizes this logic with a second-order approximation around the no-shock benchmark. It expresses mean growth, dispersion, and skewness as functions of the shock process and the local derivatives of $Q^\star$. Note that section 2 measures skewness with Kelley skewness, a robust quantile-based index which is standard in the literature. However, since it does not lend itself to tractactable approximations we study its moment-based analogue here. For any random variable X, let

$$\gamma_1[X] \equiv \frac{\mathbb{E}[(X - \mathbb{E}X)^3]}{\mathbb{V}[X]^{3/2}} \tag{6}$$

[10] We state the shocks as cost shocks for concreteness. All subsequent results also hold for shocks to the scale of the inverse demand function; such shocks generate a monopolist's problem that is isomorphic to the one studied below.

denote its standardized third central moment. Like Kelley skewness, γ_1 captures whether a distribution has a longer left or right tail; hence, both measures yield the same qualitative predictions.

Proposition 1 (Approximate Growth Rate Moments)**.** *Let all moments be finite, κ_t, λ be symmetrically distributed for all t, and $\kappa_t \overset{D}{\sim} \kappa_{t-1}$ up to a difference in variances. Let $Q_1^\star \equiv (Q^\star)'(0)$ and $Q_2^\star \equiv (Q^\star)''(0)$, and denote $\mathbb{V}[\lambda] = \sigma_\lambda^2 = 1$. Then, up to a second-order approximation:*

$$\mathbb{E}[g_t] - a \simeq \frac{Q_2^\star}{2}(\Delta\sigma_{\kappa,t}^2 + \Delta u_t^2 \sigma_\lambda^2) - b\,\overline{\lambda}\,\Delta u_t. \tag{7}$$

For the variance,

$$\mathbb{V}[g_t] \simeq (Q_1^\star)^2 \cdot \Big(\sigma_{\kappa,t}^2 + \sigma_{\kappa,t-1}^2 + (\Delta u_t)^2\sigma_\lambda^2\Big). \tag{8}$$

For the moment-based skewness measure:

$$\gamma_1(g_t) \simeq \frac{Q_2^\star}{|Q_1^\star|}\frac{3}{2}\frac{\Delta\mathbb{V}(\kappa_t^2) + 4(\Delta u_t)(u_{t-1}\sigma_{\kappa,t-1}^2 + u_t\sigma_{\kappa,t}^2)\sigma_\lambda^2 + (\Delta u_t^2)(\Delta u_t)^2\mathbb{V}[\lambda^2]}{\Big(\sigma_{\kappa,t}^2 + \sigma_{\kappa,t-1}^2 + (\Delta u_t)^2\sigma_\lambda^2\Big)^{3/2}}. \tag{9}$$

Proof of Proposition 1. We prove the proposition using a second-order Taylor approximation of g_t around 0 and then calculate the relevant moments, discarding all terms above second order. For a complete derivation, see Lemma 2 in the Appendix. All subsequent proofs are also in the Appendix. ■

Proposition 1 shows that one local object does most of the work: $Q_2^\star$. If $Q^\star$ were linear, then $Q_2^\star = 0$, so shock dispersion would matter for the variance of growth but not for the mean through Jensen effects, and it would not generate skewness at all. Concavity, $Q_2^\star < 0$, is therefore the minimum reduced-form ingredient needed to connect the first three moments of growth.

Mean and Stylized Fact 1. Equation (7) separates average growth into a direct aggregate effect and a curvature effect. The term $-b\,\overline{\lambda}\,\Delta u_t$ is the direct effect: when the contractionary aggregate factor rises, all firms face higher costs and average growth falls. The term involving $Q_2^\star$ is a Jensen effect. If $Q^\star$ is concave, an increase in shock dispersion lowers average growth because firms receiving a beneficial shock expand log output by less than firms receiving an adverse shock of equal magnitude contract. Downturns are therefore especially sharp when the aggregate shock both raises common costs and spreads firms out along a concave policy. This is the recession-side asymmetry in Stylized Fact 1.

Dispersion and Stylized Fact 2. Equation (8) shows why aggregate shocks generate the V-shape in dispersion. The aggregate factor enters through $(\Delta u_t)^2$: firms carry the same exposure from $t-1$ to t, so cross-sectional growth dispersion is driven by changes in the aggregate state, not by its level alone. A rise in u_t spreads firms out during the contraction, and a fall in u_t spreads them out again during the recovery. Both movements raise dispersion. Pure idiosyncratic uncertainty also raises dispersion, but it is less tightly linked to the sign and timing of aggregate growth.[11] This is why an aggregate-exposure channel naturally matches the clean relation between dispersion and the absolute deviation of growth from trend.

Skewness and Stylized Fact 3. Equation (9) shows that skewness is governed by the same curvature term, now normalized by the slope of the policy. At firm i's period-t centered shock realization $\epsilon_{i,t} := \kappa_{i,t} + \lambda_i u_t$, define the *local curvature index* as

$$\rho\big[Q^\star\big](\epsilon_{i,t}) := \frac{Q^{\star\prime\prime}(\epsilon_{i,t})}{|Q^{\star\prime}(\epsilon_{i,t})|}. \tag{10}$$

Thus, $\rho[Q^\star](\epsilon_{i,t}) < 0$ implies that the policy is locally concave around $\epsilon_{i,t}$. Let $\epsilon^\star$ denote the benchmark shock and define

$$\rho^\star := \rho[Q^\star](\epsilon^\star)$$

as the local policy curvature around that benchmark. Throughout the rest of the paper, we set $\epsilon^\star = 0$, the no-shock benchmark. Hence, $\rho^\star < 0$ implies that the policy is locally concave at the no-shock benchmark, with more negative values corresponding to stronger local concavity. Proposition 1 shows that $\rho^\star$ scales the moment-based skewness response. The sign and size of skewness therefore come from two objects: how much the shock spreads firms out across the policy, and how sharply the policy bends at the benchmark.

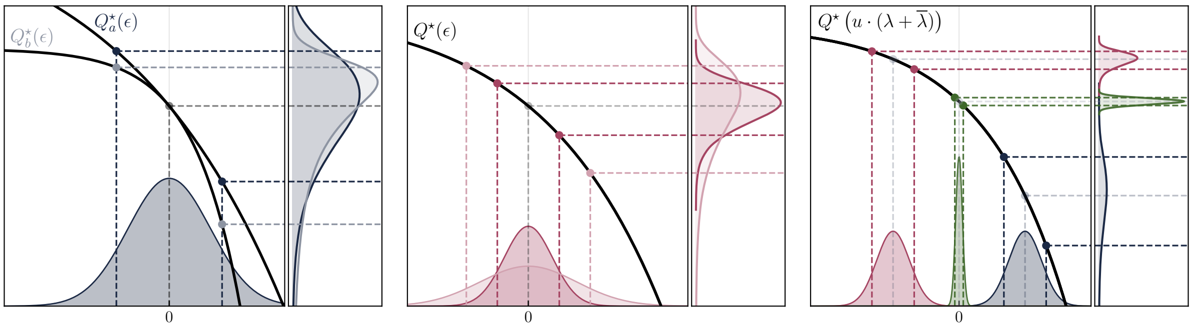
This observation gives the impact–rebound pattern in Stylized Fact 3. On impact of a contractionary aggregate shock, current shock incidence is more dispersed than lagged incidence. Local concavity then maps the current cross section into a long left tail, so growth skewness turns negative. During the recovery, the previously dispersed state is being subtracted from current log output. The sign flips, and skewness becomes positive.

The same curvature index also gives the size-gradient prediction. Holding the shock process fixed, $\rho^\star$ scales the amplitude of the skewness response. Firms with more negative local curvature therefore display larger skewness swings after the same aggregate impulse. This is our proposed channel for the size gradient of skewness and motivates the market-power microfoundation below.

Figure 4 illustrates the two ingredients in (9). Panel (a) holds the shock distribution fixed and changes the curvature of $Q^\star$: the more concave policy compresses favorable shocks more strongly and produces a more left-skewed distribution of log output. Panel (b) holds the policy fixed and increases the variance of shocks: a wider shock distribution places more mass in regions where curvature matters, amplifying left skewness.

[11] For example, consider the sequence $(\sigma_{\kappa,t-1}, \sigma_{\kappa,t}, \sigma_{\kappa,t+1}) = (0, 1, 1)$, holding $a, u = 0$ fixed throughout. A straightforward calculation shows that $\mathbb{E}[g_t] < 0 = \mathbb{E}[g_{t+1}]$ while $\mathbb{V}[g_t] < \mathbb{V}[g_{t+1}]$. This positive correlation between mean and dispersion contradicts the U-shaped relationship in Stylized Fact 2.

Figure 4: Skewness in log-Output: Concavity, Shock Variance, and Non-Homotheticity



(a) Concave policies (b) Shock variance (c) Non-homotheticity

Note: Panel (a) shows how a symmetric input (shock) distribution ϵ is mapped into skewed log-output $\widehat{q} = \ln q$ through concave production policies. Since $Q_b^\star$ is more locally concave than $Q_a^\star$, the induced distribution is more left-skewed. Panel (b) shows how increasing the variance of the input distribution amplifies skewness in outcomes. Panel (c) illustrates a non-homothetic setting in which the location of the shock distribution shifts with the aggregate state. Different colors correspond to different aggregate states; the resulting output distributions differ in location and dispersion, with dispersion responding asymmetrically to adverse versus favorable shocks. Dashed lines indicate the 10th, 50th, and 90th percentiles.

Panel (c) shows the problems that would arise if we could not pass the aggregate effect through $Q^\star$. Here, the entire population of firms evaluates its policies at a different average location after an aggregate shock. In this case we would be unable to learn from the steady-state equilibrium, and the comovement of a different firm-level dispersion and growth skewness would be undetermined from local behavior. However, homotheticity of the underlying demand system is a virtuous assumption which not only rules out this indeterminacy, but also implies desirable aggregate behavior: if every firms' cost increases by the same relative amount—or if the total market size shrinks—then market shares remain unaltered. Supplemental Appendix B formalizes this separation in a monopolistic-competition equilibrium.

These results show that cyclical skewness does not require a primitive third-moment shock. A mean-shifting aggregate shock is enough: it moves average growth, spreads firms out through heterogeneous exposure, and local concavity maps that spread into skewness. A pure uncertainty shock can also generate skewness by raising cross-sectional dispersion, but it does not tie skewness as directly to the recession and recovery in mean growth. Remark 1 makes this distinction precise.

Remark 1 (Skewness dynamics under aggregate versus uncertainty shocks)**.** *Consider an aggregate shock decaying at rate β: $(u_0, u_1, u_2, \ldots) = x \cdot (0, \beta^0, \beta^1, \ldots)$ with $x > 0$, setting idiosyncratic shocks to* 0. *Substituting into* (9)*, γ_1 scales in x, is negative on impact, and decays exponentially:*

$$(\gamma_{1,1}, \gamma_{1,2}, \gamma_{1,3}, \ldots) \propto \big((-1), \beta^0(1+\beta^{-1}), \beta^1(1+\beta^{-1}), \ldots\big).$$

This pattern coincides with the impulse response of the mean in both sign and rate of decay. Alternatively, consider a similar impulse to the standard deviation of idiosyncratic shocks. The corresponding skewness response is

$$(\gamma_{1,1},\, \gamma_{1,2},\, \gamma_{1,3},\, \ldots) \propto \Big((-1),\, \beta^0 \frac{1-\beta^2}{\sqrt{1+\beta^2}},\, \beta^1 \frac{1-\beta^2}{\sqrt{1+\beta^2}},\, \ldots\Big).$$

While the decay is also exponential, the skewness rebound is at least four times stronger for the aggregate shock than for the uncertainty shock.[12] *One would therefore expect the impulse response of skewness to an aggregate shock to display a stronger rebound than that to an uncertainty shock. The most direct comparison of this kind is between the oil and uncertainty shock IRFs in Figure 9 in the empirical section, which confirms this prediction.*

4 A Demand-Side Microfoundation

The reduced form identifies the object that matters for skewness: the benchmark curvature $\rho^\star$. This section asks where that curvature comes from. Why should a price-setting firm have $\rho^\star < 0$? And why should this curvature become more negative when the firm has more market power? The answer is demand-side markup adjustment. Price taking gives an affine log-output policy, so its benchmark curvature is zero. Under variable-elasticity demand, price-setting firms absorb favorable shocks partly through prices and markups, while adverse shocks translate more strongly into quantities. This makes the log-output policy locally concave. As the firm moves away from the price-taking benchmark, this curvature becomes more negative, so market power amplifies the skewness response to a given aggregate shock. To keep the main text focused on the local curvature mechanism, we analyze a single firm with isoelastic costs and a general inverse demand curve.[13,14]

4.1 Firm Problem and the Price-Taking Benchmark

Firms produce with convex cost

$$c(q;\epsilon) = q^\eta e^{\epsilon + C}, \tag{11}$$

where $\eta > 1$, $C > 0$ is a constant, and q denotes output. The term e^ϵ is a stochastic cost shifter. We assume that ϵ is drawn from a symmetric distribution with finite variance and

[12]For $\beta \in (0,1)$, consider $R(\beta) := \frac{(1-\beta^2)/\sqrt{1+\beta^2}}{1+1/\beta}$. One checks that $R'(\beta) = -(\beta^3+2\beta-1)/(1+\beta^2)^{3/2} = 0$ holds at $\beta \approx 0.45$, where R attains its unique maximum at $R(\beta) \approx 0.226 < 1/4$.

[13]The iso-elastic cost function isolates the competition channel. Supply-side nonlinearities can also generate skewness, but they do not naturally explain why skewness swings should be systematically larger for firms with more market power. We return to such mechanisms in Supplemental Appendix A.

[14]Supplemental Appendix B embeds the same policy representation in a homothetic monopolistic-competition equilibrium.

is observed by the firm at the time of production. The reference value of the shock is 0, so ϵ can stand in for the firm-level realization of the factor structure from Section 3. We deliberately keep the cost side iso-elastic: the paper is about demand-side nonlinearities and markup adjustment, not about mechanical one-sided technological constraints.[15]

The price-setting problem. The firm faces inverse demand $p(q)$ and solves

$$\max_{q\geq 0} \; qp(q) - c(q;\epsilon). \tag{12}$$

Assumption 1 guarantees that the problem is well behaved and the optimum is unique.

Assumption 1 (Regularity Conditions)**.** *The inverse demand function p satisfies:*

i) $p : \mathbb{R}_+ \to \mathbb{R}_+$ *and* $p \in \mathcal{C}^3(\mathbb{R}_+)$*,*

ii) $p'(q) < 0$ *on the relevant operating domain,*

iii) $\frac{\partial^2}{\partial q^2} \log p(q) \leq 0$*,*

iv) marginal revenue is positive at some finite quantity level.

The optimality condition takes a Lerner-style form. For any differentiable function f with nonzero sign, define its elasticity by

$$\mathcal{E}f(q) := \frac{qf'(q)}{f(q)}. \tag{13}$$

Lemma 1 then characterizes the solution to (12).

Lemma 1 (Solution of Firm Problem)**.** *The monopolist's solution is unique, interior, and implicitly given by*

$$\eta e^{\epsilon + C} q^{\eta - 1} = p(q)\big(1 + \mathcal{E}p(q)\big) =: \mathrm{mr}(q), \tag{14}$$

$$\iff \epsilon = \log \mathrm{mr}(e^{\widehat{q}}) - \log \eta - (\eta - 1)\widehat{q} - C =: H(\widehat{q}), \qquad \widehat{q} := \log q. \tag{15}$$

Optimal log output, $Q^\star(\epsilon)$, is decreasing in ϵ. On the relevant domain, inverse-demand elasticity satisfies $\mathcal{E}p(q) \in (-1, 0)$ and the markup is $\mu(q) = \big(1 + \mathcal{E}p(q)\big)^{-1}$.

Lemma 1 gives us a convenient way to study the curvature of the output policy. The first-order condition defines $H(\widehat{q})$, which is the inverse of $Q^\star(\epsilon)$. We can therefore study the curvature of $Q^\star$ by studying the curvature of H, or equivalently the geometry of marginal revenue. If H is affine, then $Q^\star$ is affine and symmetric shocks do not generate skewness. Demand matters only insofar as the marginal-revenue schedule bends this affine benchmark.

[15] All subsequent results also hold for shocks to the scale of inverse demand, which generate an isomorphic firm problem.

The price-taking benchmark. The price-taking case is exactly this affine benchmark. A price-taking firm chooses output taking the market price $\bar{p}$ as given. Its first-order condition is

$$\epsilon = \log \bar{p} - \log \eta - (\eta - 1)\widehat{q}_{\mathrm{pt}} - C. \tag{16}$$

The right-hand side is affine in log output, so the inverse policy $Q^{\star}_{\mathrm{pt}}(\epsilon)$ is affine in the shock. With no markup-adjustment margin, its benchmark curvature is $\rho^{\star}_{\mathrm{pt}} := \rho[Q^{\star}_{\mathrm{pt}}](\epsilon^{\star}) = 0$. Price taking therefore provides the zero-curvature benchmark against which we measure the effects of market power.

4.2 From Demand Shape to Local Curvature

Section 3 showed that the mapping from aggregate first-moment shocks to skewness runs through one object: the benchmark curvature $\rho^{\star}$. This subsection asks when demand primitives make that index negative. We hold the inverse demand curve fixed and derive local restrictions under which a price-setting firm's log-output policy is concave around the no-shock allocation. These restrictions connect the primitive shape of demand to the moment implications of the reduced form.

Because Lemma 1 writes the policy through the inverse map H, the sign of $\rho^{\star}$ can be read from the local shape of marginal revenue. Recall that $\epsilon^{\star} = 0$ and let

$$q^{\star} := \exp\bigl(Q^{\star}(\epsilon^{\star})\bigr) \tag{17}$$

denote the corresponding reference quantity. All demand restrictions below are evaluated locally at this reference quantity.

Demand-shape objects. To characterize the shape of marginal revenue in economic terms, we use the elasticity of inverse demand and its local derivatives:

$$\varepsilon(q) := \mathcal{E}p(q) < 0, \qquad \psi(q) := \mathcal{E}\varepsilon(q) = \frac{q\,\varepsilon'(q)}{\varepsilon(q)}, \qquad \Pi(q) := q\,\psi'(q). \tag{18}$$

Here ε is the elasticity of inverse demand, ψ is its *superelasticity* (Melitz, 2018), and Π is the slope of that superelasticity. At the reference quantity, write $\varepsilon^{\star} := \varepsilon(q^{\star})$, $\psi^{\star} := \psi(q^{\star})$, and $\Pi^{\star} := \Pi(q^{\star})$. The restrictions below use these objects to link demand primitives to policy curvature. We first state a direct condition on marginal revenue, then connect it to inverse-demand curvature, and finally relate that curvature to pass-through, which is closer to observed firm behavior.

Property 1 (Marshallian second-law restrictions)**.**

i) Marshall's Second Law of Demand (MSLD) holds at $q^{\star}$ if the absolute elasticity of inverse demand is locally increasing, i.e. if $-\varepsilon'(q^{\star}) > 0$.

ii) Marshall's Strong Second Law of Demand (MSLD′) holds at $q^\star$ if the absolute elasticity of marginal revenue is locally increasing:

$$-\frac{\partial}{\partial q}\mathcal{E}\mathrm{mr}(q^\star) > 0.$$

MSLD says that as the firm expands along its demand curve, demand becomes more elastic. In variable-markup models this is the key restriction that makes product-market conditions matter for markup and reallocation responses (Mrázová and Neary, 2017; Melitz, 2018). MSLD′ is the corresponding condition on marginal revenue. It implies MSLD and, through equation (15), directly signs the curvature of $Q^\star$.

Property 2 (ISID)**.** *Increasing superelasticity of inverse demand (ISID) holds locally at the reference quantity if $\psi'(q^\star) > 0$. (Equivalently, $\Pi^\star > 0$.)*

ISID is a higher-order version of MSLD. Economically, it says that as the firm moves to higher quantities, elasticity itself becomes more sensitive to further quantity changes. This is harder to interpret and to measure directly than MSLD, which is why it is useful to connect it to pass-through.

Property 3 (IPT)**.** *Let $\bar{c}^\star$ denote the cost level that supports the reference quantity in the auxiliary problem below; for the firm problem above, $\bar{c}^\star = \exp(\epsilon^\star + C)$. For any scalar cost level $\bar{c} > 0$, let $q(\bar{c})$ solve $\max_{q\geq 0}\ p(q)q - \bar{c}\,q^\eta$ and let $p^{\mathrm{opt}}(\bar{c}) := p(q(\bar{c}))$. We say that increasing pass-through (IPT) holds locally at the reference quantity if the level pass-through*

$$\tau(\bar{c}) := \frac{dp^{\mathrm{opt}}(\bar{c})}{d\bar{c}} \tag{19}$$

is strictly increasing at $\bar{c}^\star$.

Pass-through is closer to observed firm behavior than the slope of superelasticity. Here τ is level pass-through: it measures the change in the firm's price level induced by a marginal increase in the multiplicative cost shifter that scales the entire cost function. Thus IPT says that a marginal increase in the cost shifter raises the firm's price by more when the firm is already in a higher-cost state.[16] Since pass-through can be estimated from price responses, it can be easier to bring to the data than ψ or Π themselves (Weyl and Fabinger, 2013; Amiti et al., 2019).

[16] A familiar empirical pass-through rate is log-log pass-through, $\theta(\bar{c}) := d\log p^{\mathrm{opt}}(\bar{c})/d\log\bar{c}$. For example, if the cost function is scaled up by a factor of 1.1 and the firm's price rises by 8%, then $\theta \simeq \log(1.08)/\log(1.1) \simeq 0.81$. The level pass-through used in the proposition is related to this rate by

$$\theta(\bar{c}) = \frac{\bar{c}}{p^{\mathrm{opt}}(\bar{c})}\tau(\bar{c}), \qquad \tau'(\bar{c}) = \frac{p^{\mathrm{opt}}(\bar{c})}{\bar{c}}\left[\theta'(\bar{c}) + \frac{\theta(\bar{c})(\theta(\bar{c})-1)}{\bar{c}}\right].$$

Hence increasing log-log pass-through rates are not sufficient for level IPT in general, but they are nevertheless useful evidence for the condition; with incomplete pass-through, $\theta(\bar{c}) < 1$, the second term can be negative.

Implications for curvature and skewness. With these definitions, let weak local ISID mean $\Pi^\star \geq 0$. Proposition 2 is the main result for a fixed inverse demand curve. It shows how demand restrictions determine the object that matters for the growth distribution, $\rho^\star$. The direct condition is MSLD′: when the absolute elasticity of marginal revenue rises locally, the firm's log-output policy is locally concave. MSLD plus weak ISID delivers this marginal-revenue condition, while IPT plus MSLD gives a pass-through route to ISID. Thus familiar variable-markup restrictions are sufficient for $\rho^\star < 0$.

Proposition 2 (Local demand-shape implications for a given inverse demand curve)**.** *At the given inverse demand curve, the following implications hold at $q^\star$:*

$$\textit{local MSLD} \ \wedge \ \textit{weak local ISID} \implies \textit{local MSLD}' \iff \rho^\star < 0, \tag{20}$$
$$\textit{local IPT} \ \wedge \ \textit{local MSLD} \implies \textit{local ISID}. \tag{21}$$

Hence local IPT together with local MSLD is sufficient for local concavity of the policy.

The economics are straightforward. A favorable cost shock moves the firm to a higher-quantity region of demand. Under MSLD′, marginal revenue is more elastic there, so the firm absorbs part of the favorable shock through a lower markup rather than a proportional increase in quantity. An adverse cost shock moves the firm toward a lower-quantity region, where quantity responds more steeply. Thus good shocks raise log output by less than equally sized bad shocks reduce it. This asymmetry is local concavity of $Q^\star$.

Corollary 1. *The price-taking policy is affine and therefore satisfies*

$$Q^{\star\prime\prime}_{\text{pt}}(\epsilon^\star) = 0, \qquad \rho^\star_{\text{pt}} = 0. \tag{22}$$

By contrast, any price-setting firm satisfying MSLD′ at $\epsilon^\star = 0$ has $\rho^\star < 0$.

Corollary 1 gives the price-taker comparison. With no markup margin, the policy is linear. A symmetric cross section of shocks therefore remains symmetric after being mapped into log output. Once the firm can move along a non-CES inverse demand curve and local MSLD′ holds, price and markup adjustment no longer move one-for-one with quantities, so the same symmetric shocks generate an asymmetric output response.

To translate this demand restriction into skewness dynamics, suppose the assumptions of Proposition 1 hold, there are no idiosyncratic shocks, and the aggregate factor follows the mean-reverting path

$$(u_0, u_1, u_2, \ldots) = x \cdot (0, \beta^0, \beta^1, \beta^2, \ldots), \qquad x > 0, \quad \beta \in (0, 1). \tag{23}$$

Combining Proposition 2 with the reduced-form moment approximation yields the following result.

Corollary 2 (Local impact and rebound of skewness)**.** *If local MSLD′ holds at the no-shock benchmark, then moment-based skewness is negative on impact and positive during* *the recovery, with monotone decay:*

$$\gamma_1(g_1) < 0 < \gamma_1(g_t) \qquad \forall\, t > 1, \qquad |\gamma_1(g_{t+1})| = \beta\, |\gamma_1(g_t)| \qquad \forall\, t \geq 2. \tag{24}$$

The amplitude of the entire path is proportional to $|\rho^\star|$.

Combining the demand-side curvature result with the reduced-form approximation gives the impact–rebound pattern. On impact, the recessionary impulse compresses the right tail of growth rates more than the left, and skewness turns negative. During the recovery, the sign flips and the magnitude decays geometrically with the shock. A pure uncertainty shock that temporarily raises idiosyncratic dispersion can generate skewness through the same concavity, but the aggregate shock ties the skewness path directly to the recession and recovery in mean growth.

4.3 From Market Power to the Size Gradient

The previous subsection explained why a price-setting firm can turn symmetric shocks into skewed output responses. The size gradient requires more: holding the shock process fixed, firms with more market power must produce larger asymmetries. This subsection establishes that result locally around the price-taking benchmark by showing that market power makes the relevant curvature index more negative.

Market-power parameterization. We parameterize market power by α in a way that nests price taking and price setting. At $\alpha = 0$, the firm takes the fixed benchmark price $\bar{p}$ as given. At $\alpha = 1$, the firm faces the inverse demand curve p studied above. For intermediate values, α scales the slope of log inverse demand, and hence the inverse-demand elasticity relevant for markups. We write

$$p_\alpha(q) := \bar{p}^{1-\alpha} p(q)^\alpha, \qquad \alpha \in [0, 1], \tag{25}$$

This specification nests the price-taking benchmark and the full price-setting problem under p.[17]

The corresponding demand-shape coefficients satisfy $\varepsilon_\alpha = \alpha\varepsilon$, $\psi_\alpha = \psi$, and $\Pi_\alpha = \Pi$. Thus, for any $\alpha > 0$, the demand restrictions from the previous subsection can be evaluated on the endpoint demand curve p: MSLD, MSLD′, and ISID hold for p_α if and only if they hold for p.[18] This matters because varying α changes the strength of the markup margin while preserving the relevant demand-shape restrictions.

[17] Appendix Lemma 5 explains why the benchmark must be a constant price: with a nonconstant benchmark demand curve, $\alpha = 0$ need not be price taking, and the markup response to α is not pinned down without additional restrictions.

[18] MSLD and ISID are immediate. For MSLD′, see Lemma 4. Pass-through is not included in this equivalence because it is defined through the firm's cost-shifter comparative static, not only through the demand-shape coefficients ε, ψ, and Π.

Equilibrium local curvature. To obtain the size-gradient comparison, we now ask how this markup margin changes the curvature of the equilibrium policy. Absorbing the constant cost shifter C into $\bar{p}$ and p, the log first-order condition for a firm with market power α facing shock ϵ can be written as

$$H_\alpha(\widehat{q}) := (1-\alpha)\log\bar{p} + \alpha\log p(e^{\widehat{q}}) + \log\big(1+\alpha\varepsilon(e^{\widehat{q}})\big) - \log\eta - (\eta-1)\widehat{q} = \epsilon. \tag{26}$$

For each $\alpha \geq 0$, let $\widehat{q}_\alpha^\star$ solve $H_\alpha(\widehat{q}_\alpha^\star) = \epsilon^\star$, where, as throughout, $\epsilon^\star = 0$ is the no-shock benchmark. The benchmark curvature associated with policy $Q_\alpha^\star$ is

$$\rho_\alpha^\star := \rho\big[Q_\alpha^\star\big](\epsilon^\star) = \frac{Q_\alpha^{\star\prime\prime}(\epsilon^\star)}{|Q_\alpha^{\star\prime}(\epsilon^\star)|} = \frac{H_\alpha''(\widehat{q}_\alpha^\star)}{H_\alpha'(\widehat{q}_\alpha^\star)^2}. \tag{27}$$

This notation extends the fixed-policy objects above: $\rho_1^\star = \rho^\star$ and $\rho_0^\star = \rho_{\mathrm{pt}}^\star = 0$. The size-gradient comparison asks whether a small move away from price taking pushes this curvature below zero. Let $q_0^\star = \exp\widehat{q}_0^\star$ denote the competitive reference quantity; the demand restrictions in the next proposition are evaluated there.

Proposition 3 (Sufficient conditions for one-sided monotone curvature)**.** *Let $\varepsilon(q_0^\star)$, $\psi(q_0^\star)$, and $\Pi(q_0^\star)$ be the elasticity, superelasticity, and slope of superelasticity of p evaluated at $q_0^\star$.*

i) For small α, the curvature response at the competitive benchmark is

$$\left.\frac{d\rho_\alpha^\star}{d\alpha}\right|_{0+} = \frac{\varepsilon(q_0^\star)}{(\eta-1)^2}\Big(\psi(q_0^\star) + \psi(q_0^\star)^2 + \Pi(q_0^\star)\Big). \tag{28}$$

ii) If local MSLD and weak local ISID hold for p at $q_0^\star$, then

$$\left.\frac{d\rho_\alpha^\star}{d\alpha}\right|_{0+} < 0. \tag{29}$$

Moreover, because local level IPT together with local MSLD for p implies local ISID for p, local IPT and local MSLD for p are also sufficient for the one-sided inequality in (29).

Part i) is the key comparative static for the size gradient: it gives the derivative of equilibrium local curvature with respect to market power at $\alpha = 0$. Because $\varepsilon(q_0^\star) < 0$, the sign of the curvature response is governed by the superelasticity term in brackets. The demand-shape conditions in part ii) make that bracket positive, so turning on market power makes $\rho_\alpha^\star$ more negative. The pass-through clause gives an empirically closer sufficient route: local level IPT together with MSLD for the endpoint demand curve p implies ISID for p, and therefore implies the same one-sided curvature result.

The economics are those of state-dependent markup adjustment. Once a firm has pricing power, favorable shocks are absorbed relatively more through price and markup adjustment and relatively less through quantities, whereas unfavorable shocks still translate into relatively sharp contractions. That is why the policy bends downward as market power rises.

Implications for skewness. To verify the conditions of Proposition 3, one does not need to recover the latent decomposition into the benchmark price $\bar{p}$ and the endpoint demand curve p. For firms with positive but sufficiently small α, the relevant local coefficients can be measured on the demand curve p_α that the firm actually faces. This curvature ranking closes the loop back to the growth distribution. Corollary 3 applies the reduced-form moment result to show that the same shock process generates the same sign pattern of skewness dynamics across firms, but larger amplitudes for firms with more market power.

Corollary 3. *Suppose the conditions of Proposition 1 hold and firms differ in market power only through α. If local MSLD and weak local ISID hold for $p_{\widetilde{\alpha}}$ at $q_0^\star$ for some (equivalently, any) $\widetilde{\alpha} > 0$ on the path, then there exists $\bar{\alpha} > 0$ such that*

$$0 < \alpha_1 < \alpha_2 < \bar{\alpha} \qquad \Longrightarrow \qquad \rho^\star_{\alpha_2} < \rho^\star_{\alpha_1} < 0. \tag{30}$$

Hence the skewness response is procyclical and its amplitude is locally increasing in market power:

$$|\gamma_1(g_{t,\alpha_2})| > |\gamma_1(g_{t,\alpha_1})| \tag{31}$$

for every horizon at which the shock term in Proposition 1 is nonzero.

The amplification mechanism therefore runs entirely through local curvature. The model does not need exogenous skewness shocks, nor does it need ad hoc asymmetries on the production side. Once firms differ in market power, standard demand-side restrictions are enough to generate procyclical skewness and a size gradient in its amplitude.

4.4 Discussion

Economic intuition. The core object in this section is the curvature of marginal revenue around the firm's reference quantity. Under price taking, or under CES demand, that object is flat and the policy is locally linear. Symmetric shocks then stay symmetric in growth rates. Under variable-elasticity demand, markup adjustment becomes state dependent. Expansions are absorbed relatively more through prices and markups and relatively less through quantities, while contractions still generate sharp quantity declines. The firm policy bends downward, and a symmetric cross section of shocks turns into a left-skewed cross section of outcomes.

How these demand objects enter the literature. The paper builds on the variable-elasticity monopolistic-competition literature that generalizes CES while preserving tractability. Feenstra (2003) and Parenti et al. (2017) show that homothetic monopolistic competition need not rely on constant elasticity demand. Mrázová and Neary (2017) show that elasticity and curvature objects provide sufficient statistics for many firm-side comparative statics, and Melitz (2018) emphasizes Marshall's laws and related shape restrictions as the determinants of markup and reallocation responses to tougher competition. Weyl and Fabinger (2013) explain why pass-through is a particularly useful reduced-form object under imperfect competition. Matsuyama and Ushchev (2022) push the same logic into selection and sorting, showing how the second and third laws govern markup and pass-through responses to competitive pressure.

What is the empirical support for MSLD, IPT, and related objects? Direct evidence on ISID itself is understandably scarce, because it is a higher-order derivative of demand rather than an object that firms or researchers observe directly. Empirical work therefore proceeds indirectly. Mayer et al. (2021) document that positive demand shocks shift French firms' product mix toward their better-performing products and show that the demand conditions needed for those reallocations imply MSLD. Berman et al. (2012) show that high-performance exporters react to exchange-rate movements much more through markup adjustment and much less through quantities than low-performance exporters. Amiti et al. (2019) directly estimate firm-level price responses and find strong strategic complementarities and incomplete pass-through for larger firms, whereas small firms are close to complete pass-through and show little response to competitors' prices. Taken together, these findings support the view that large firms operate on variable-elasticity demand curves and far from the price-taking benchmark. Consistent with that interpretation, firm-level markups rise with export status and broader "superstar firm" evidence links concentration to higher markups and market power (De Loecker and Warzynski, 2012; Autor et al., 2020).

Benchmarks. The theory does not rely on exotic demand systems. Linear inverse demand satisfies the one-sided monotone-curvature result at the competitive benchmark. By contrast, CES is the knife-edge case in which superelasticity is identically zero, the policy remains locally linear for every α, and skewness is absent to second order; see Proposition 4 in the Appendix. The point is not that any deviation from CES will do, but that economically familiar deviations from CES generate exactly the curvature needed for the data.

Connection to the stylized facts. Put together, the theory gives a tight reading of the evidence documented earlier in the paper. An adverse aggregate shock with heterogeneous exposure explains why mean growth falls sharply and why dispersion rises whenever growth moves far from trend (Stylized Facts 1 and 2). Local demand-side concavity explains why skewness moves with the mean, and the market-power comparative static explains why those skewness swings are larger for large firms and for firms sorted by profitability, estimated markups, or industry concentration (Stylized Fact 3). The empirical section takes these predictions to the data by comparing firms and industries with different market-power measures, exploiting heterogeneity in pandemic exposure, and tracing skewness responses to identified aggregate shocks.

Why demand rather than production? None of this rules out supply-side sources of skewness. Capacity constraints or other one-sided adjustment frictions can make production policies locally concave and thereby generate asymmetric growth responses. The difficulty is explaining the cross-sectional gradient: a production-side account would have to explain why policy curvature is systematically stronger among larger, more profitable, and higher-markup firms and in more concentrated industries. This is not an obvious technological primitive. Demand curvature provides a more direct explanation because market power varies persistently along these same firm and industry characteristics, and the model links that variation to policy curvature through markup adjustment. The distinctive contribution of the demand-side microfoundation is therefore not merely to generate skewness, but to explain why its cyclical amplitude rises with the market-power measures observed in the data.

5 Empirical Evidence

Having established the theoretical link between heterogeneous exposures, market power and procyclical skewness, we now test these predictions empirically. Using quarterly Compustat data on US public firms, we first describe the data and examine how skewness varies with several market-power measures. We then exploit heterogeneous pandemic exposures as a natural experiment and estimate impulse responses to a broad set of identified aggregate shocks.

5.1 Data

Our analysis uses quarterly data on US public firms from Compustat between 1983 and 2024, covering more than 40 years of data.[19] The long sample period enables us to cover multiple recessions and draw general conclusions about growth dynamics over US business cycles. We focus, as before, on sales. Let $q_{i,t}$ be firm i's real sales in quarter t.

[19] Details on sample construction, variable definitions, and data source can be found in Appendix C. The data cleaning filters out roughly a quarter of the observations from raw Compustat. Despite this restriction, the time series of cross-sectional skewness is very similar before and after cleaning (Figure A.2), and Appendix D confirms that procyclical skewness is robust across skewness measures and cleaning procedures.

Table 1: Summary Statistics for Compustat Data

	Full Compustat	Cleaned Sample	QFR
Assets (mln. USD)	4,005	1,884	43.2
Sales (mln. USD)	400.2	414.8	10.8
Sales Growth (%)	7.2	7.6	0.63
Q(Sales Growth)$_{0.25}$ (%)	-7.8	-7.5	-25.3
Q(Sales Growth)$_{0.75}$ (%)	20.7	21.1	26.6
Net Leverage (%)	26.9	12.4	20.0
Short-term debt (%)	75.9	8.7	33.0
Obs./quarter	6,337	4,828	6,122
Unique firms	22,103	17,327	–
ρ(Sales Gr., GDP Gr.)	0.64	0.55	–
ρ(Sales Gr., Skew)	0.85	0.88	–

Note: Statistics for QFR are for the manufacturing subset of Crouzet and Mehrotra (2020) from 1977Q3–2014Q1 and directly taken from tables 1 and 3 of their paper; the values are unweighted averages across size bins. The Compustat statistics are for 1983Q3–2014Q1. The reported values for assets and sales are in 2009 USD. Compustat values are deflated using the GDP deflator (GDPDEF), rebased to 2009.

Year-on-year real sales growth is $g_{i,t} = \ln(q_{i,t}/q_{i,t-4})$. Our main business cycle indicator is aggregate real sales growth, constructed as the size-weighted average of existing firms' growth rates:[20]

$$g_t = \frac{\sum_i g_{i,t} q_{i,t-4}}{\sum_i q_{i,t-4}}.$$

This definition only considers firms present in both t and $t-4$, abstracting from entry and exit dynamics. Our main skewness measure is the Kelley skewness as defined in equation (1), applied to the empirical CDF of growth rates at each date t.

In order to put the firm characteristics obtained from Compustat into perspective, Table 1 compares the full Compustat sample against our cleaned data, together with summary statistics from the Quarterly Financial Reports (QFR), which Crouzet and Mehrotra (2020) use to construct a representative sample of US manufacturing firms. The average Compustat firm is considerably larger than the average QFR firm, both in assets and sales. Furthermore, the Compustat sales growth distribution has a larger mean, lower variance and is more positively skewed. Comparing the full Compustat sample to the cleaned version, the correlation between skewness and aggregate sales growth is virtually identical across both Compustat samples (0.85 vs. 0.88). Note, finally, since all firms in the sample are large relative to the universe of US firms, extensive-margin movements are limited and any bias from firm exit should be negligible.

[20] We treat production and sales as equal, hence ignoring inventories.

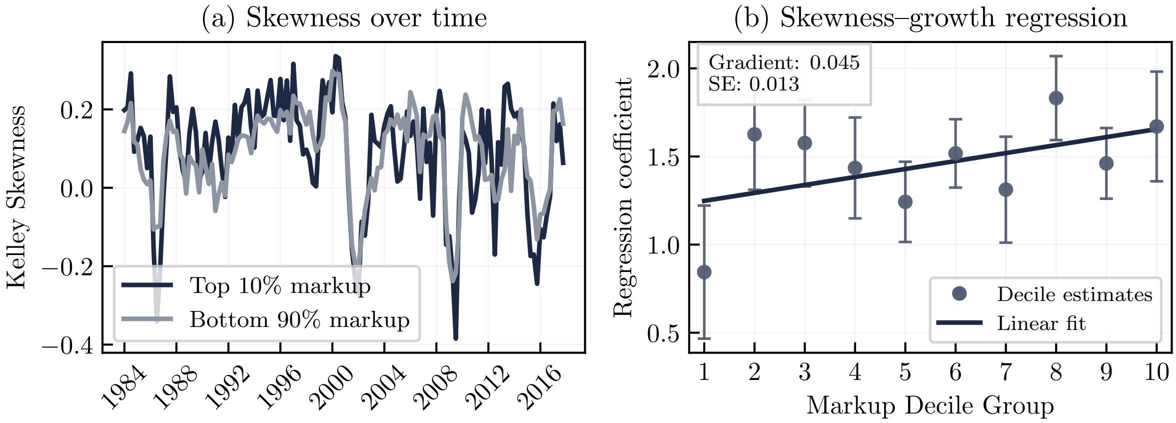


Figure 5: Estimated Markups and the Cyclicality of Skewness

Note: The sample spans 1984Q1–2017Q4. Firms are ranked using average log markups from the preceding three years. Panel (a) presents Kelley skewness for high- and low-markup firms. Panel (b) reports the sensitivity of decile-level skewness to aggregate sales growth; quarter-decile cells contain at least 10 firms. Inference uses Newey–West standard errors with four quarterly lags.

5.2 Market Power and Skewness

The size gradient is suggestive of our market-power mechanism, given the close empirical relationship between firm size and market power (Autor et al., 2020). To assess this interpretation more directly, we turn to estimated markups and examine whether the same pattern is also present in firm profitability and industry concentration. These measures capture different dimensions of market power and are subject to different measurement concerns, but they point in the same direction: skewness is generally more cyclical among firms and industries associated with greater market power.

Estimated markups. We begin with the proxy conceptually closest to market power. We construct firm-level markups following De Loecker et al. (2020), using the full-sample implementation of Benkard et al. (2025). Building on De Loecker and Warzynski (2012), these estimates divide the estimated production-function elasticity with respect to a composite variable-input bundle by that bundle's expenditure share in revenue. Following the Compustat implementation of De Loecker et al. (2020), cost of goods sold proxies total expenditure on the variable-input bundle.

The annual markup data cover 1955–2016. Because firms are ranked using markups from the preceding three years, the 2016 observations can still be used to classify firms in 2017; after matching to our sales-growth data, the quarterly estimation sample therefore spans 1984Q1–2017Q4. To limit the influence of measurement noise, we remove within each year the 1 percent of observations with the largest absolute change in log markups. As in the size-gradient analysis in Section 2, we then calculate quarterly Kelley skewness within each markup decile and estimate a pooled linear gradient in its sensitivity to aggregate growth across decile ranks.

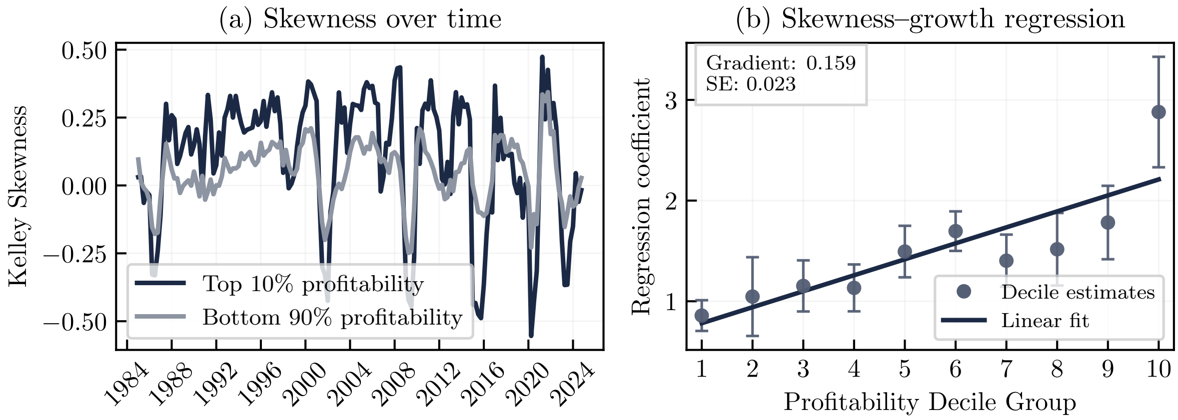


Figure 6: Profitability and the Cyclicality of Skewness

Note: Firms are ranked using average operating profitability over the preceding three years. Panel (a) presents Kelley skewness for high- and low-profitability firms. Panel (b) reports the sensitivity of decile-level skewness to aggregate sales growth.

Figure 5 is then the markup counterpart to the size analysis in Figure 3. Panel (a) shows that skewness among high-markup firms fluctuates more strongly over the business cycle than among lower-markup firms. Panel (b) confirms that this difference is not confined to the top decile: the skewness-growth coefficient generally rises across the markup distribution. The fitted gradient is positive and statistically significant at 0.045 per decile, with a standard error of 0.013, consistent with the theory's prediction that skewness cyclicality increases with market power.[21]

Production-based markup estimates nevertheless require caution. In Compustat-style data, they are constructed using revenue rather than physical output. Under imperfect competition, they may therefore combine variation in market power with demand-induced price movements (Bond et al., 2021). Consistent with this concern, Ridder et al. (2026) finds only modest pooled correlations between revenue- and quantity-based markup estimates, although the relationship is stronger within sectors. We therefore complement the markup results with two additional proxies for market power: firm profitability and industry concentration.

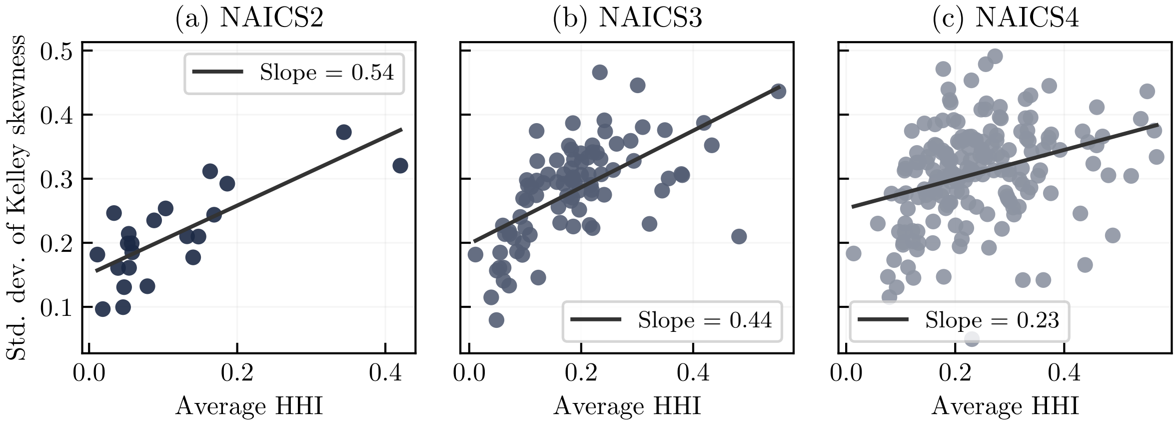
Profitability. We next use operating profitability as another firm-level proxy for market power:

$$\text{prof}_{i,t} = \frac{\texttt{saleq}_{i,t} - \texttt{cogsq}_{i,t} - \texttt{xsgaq}_{i,t}}{\texttt{saleq}_{i,t}}.$$

Mirroring the size-based analysis, firms are ranked each quarter using their average profitability over the preceding three years. Figure 6 shows substantially larger cyclical swings in skewness among highly profitable firms in Panel (a), while Panel (b) shows a clear in-

[21]The positive markup gradient survives alternative industry exclusions and specifications that rank firms within two-digit NAICS industries.

Figure 7: Skewness Variation and Industry Concentration



Note: The figure relates an industry's average HHI to the time-series standard deviation of Kelley skewness of firm sales growth. Panels (a)–(c) use two-, three-, and four-digit NAICS industries, respectively. Each observation represents an industry.

crease in the skewness-growth coefficient across profitability deciles. The fitted gradient is positive and statistically significant at 0.159 per decile, with a standard error of 0.023, supporting the prediction that skewness cyclicality is stronger among firms with greater market power.

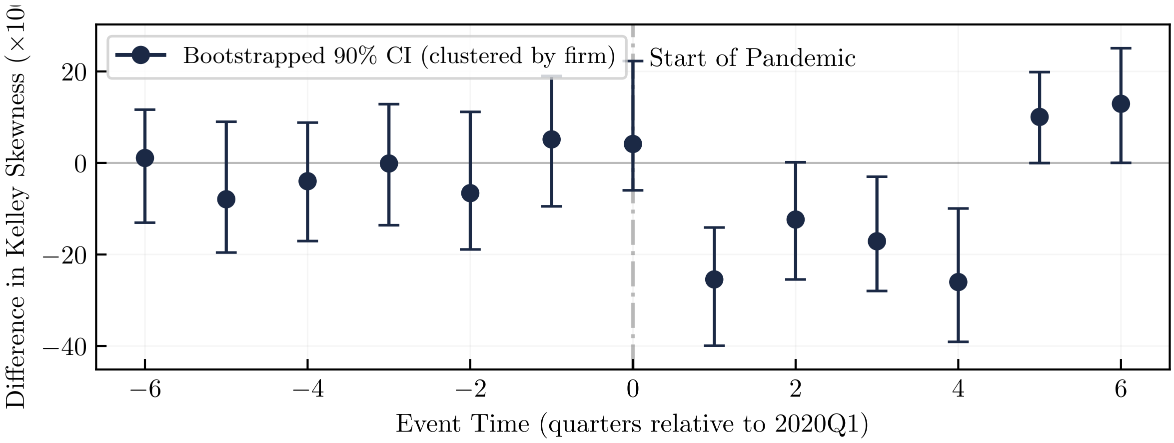
Industry concentration. Finally, we exploit variation in concentration across industries. In our framework, a higher HHI signals a larger role for firms with market power, whose output responds more asymmetrically to shocks. Holding the distribution of shock exposures fixed, more concentrated industries should therefore exhibit larger cyclical swings in firm-growth skewness. For each two-, three-, and four-digit NAICS industry, we compute the time-series standard deviation of Kelley skewness and relate it to the industry's average HHI.

Figure 7 shows a positive relationship at each level of aggregation. The estimated slope declines from 0.54 for two-digit industries to 0.44 and 0.23 for three- and four-digit industries. This attenuation is consistent with the role of heterogeneous exposures in the theory. Firms within narrowly defined industries tend to be exposed more similarly to aggregate shocks, leaving less cross-sectional variation from which movements in skewness can arise. The concentration gradient is therefore strongest for broader industry groups, where exposure heterogeneity is greater.

5.3 Skewness Responses by Cross-Sectional Exposure

Market power is one source of amplification, but the theory highlights another: firms differ in how strongly they are exposed to the same aggregate shock. Holding policy curvature fixed, greater heterogeneity in exposure widens the cross-sectional distribution

Figure 8: Differential skewness response around the pandemic



Note: The figure plots event-time impulse responses of the difference in Kelley skewness of sales growth between high- and low-pandemic-exposure firms. Event-time dummies are relative to 2020Q1. Confidence bands are 90% bootstrap intervals based on resampling firms. Pandemic exposure measures come from Davis et al. (2025), to whom we are grateful for sharing their data.

of firm responses, which in turn amplifies movements in skewness. Proposition 1 therefore predicts that firms in the tails of the exposure distribution should exhibit a sharper decline in skewness when an adverse shock hits and a stronger rebound during the recovery.

We test this prediction using heterogeneity in firms' exposure to pandemic risk. As an empirical proxy for $\lambda_{i,l}$ with respect to COVID-related disruptions, we use the pandemic exposure measure of Davis et al. (2025).[22] This measure combines information from firms' pre-2020 10-K risk disclosures with equity-market reactions on pandemic jump dates to identify firm-level exposure to pandemic shocks. A key feature of this measure is that exposure is signed: some firms are adversely exposed, while others benefit from pandemic-related disruptions. To capture firms with the strongest sensitivity in either direction, we define the high-exposure group as firms in the bottom and top quartiles of the pandemic exposure distribution z_{pand}. The low-exposure group consists of firms in the middle 50 percent. This tail-versus-middle split maps directly into the theory, which predicts that cross-sectional skewness should react most for the subset of firms with large absolute exposures.

For each group, we compute quarterly Kelley skewness of sales growth and then take the difference between the high-exposure and low-exposure groups. We study the evolution of this difference in an event-time specification centered on 2020Q1, using event-time indicators to trace the dynamics around the onset of the pandemic. Inference is based on bootstrap confidence intervals obtained by resampling firms.

Figure 8 displays the results. Prior to the pandemic, skewness evolves similarly for high- and low-exposure firms. In the subsequent quarters, however, high-exposure firms experience a larger decline in skewness than low-exposure firms. This differential persists

[22] We are grateful to the authors for generously sharing these estimates with us.

for roughly four quarters and then reverses, in line with the impact-and-recovery dynamics emphasized by the theory. Overall, both the timing and the magnitude of the response support the mechanism: firms with greater exposure to an aggregate shock exhibit larger movements in cross-sectional skewness.

5.4 Growth–Skewness Comovement Following Aggregate Shocks

The pandemic exercise isolates heterogeneity in firms' exposure to a single aggregate disturbance. We now broaden the analysis to several identified shocks. Generalizing the factor structure in equation (5), firm-level growth can be written as

$$g_{i,t} = a + \Delta Q^{\star}_{\alpha_i}\left(\kappa_{i,t} + \boldsymbol{\lambda}_i^T \boldsymbol{u}_t\right) - b\,\bar{\boldsymbol{\lambda}}^T \Delta \boldsymbol{u}_t + w_{i,t}, \tag{32}$$

where $Q^{\star}_{\alpha_i}$ denotes the log-output policy of a firm with market power α_i, the vectors $\boldsymbol{\lambda}_i$ and $\boldsymbol{u}_t$ collect firm-specific factor exposures and aggregate shocks of potentially different nature, $\bar{\boldsymbol{\lambda}}$ is the cross-sectional mean of exposures, and $w_{i,t}$ is a zero-mean i.i.d. disturbance.[23] This representation maps the theory directly into data. Proposition 1 provides the moment formulas, Corollary 2 implies that contractionary aggregate shocks generate negative cross-sectional skewness on impact and positive skewness during the recovery when the local curvature index is negative, and Corollary 3 implies that this skewness response is locally stronger for firms with greater market power, provided the sufficient conditions hold. The model therefore delivers two empirical predictions: aggregate shocks should generate tight comovement between mean growth and skewness, and this comovement should be systematically stronger among large firms, which operate with greater market power (higher α).

Empirical framework. We test these predictions in an impulse-response framework by estimating the responses of skewness and growth to six identified aggregate shocks: monetary, oil, credit, uncertainty, sentiment, and TFP shocks. These shocks differ in economic origin, identification, and sample coverage, allowing us to assess whether the predicted growth–skewness comovement is robust across a broad set of aggregate shocks. Specifically, we estimate these responses using local projections (Jordà, 2005):

$$y_{t+h} = \alpha_h + \beta_h \text{shock}_t + \sum_{\ell=1}^{L} \gamma'_{\ell,h} \text{controls}_{t-\ell} + e_{t+h}, \tag{33}$$

for horizons $h = 0, \ldots, 10$ and with up to $L = 2$ lags of controls. The coefficients β_h trace out the impulse response at each horizon. The outcome variable y is either cross-sectional skewness or aggregate sales growth, computed either for the full sample or separately for

[23]The trend a can be motivated from the secular growth component discussed in Section 3, which turns into an aggregate trend in outcomes if demand systems are homothetic.

Table 2: Local projection specifications

Shock	Reference	Controls (lagged)	Sample period
Monetary	Bu et al. (2021)	Real GDP, GDP deflator, Shadow Rate, EBP	1994Q1 – 2019Q4
Oil	Baumeister and Hamilton (2019)	Real GDP, GDP deflator, Oil price	1983Q3 – 2019Q4
Credit	Gilchrist and Zakrajšek (2012)	Real GDP, GDP deflator, EBP	1983Q3 – 2019Q4
Uncertainty	Ludvigson et al. (2021)	Real GDP, GDP deflator, VXO	1983Q1 – 2015Q4
Sentiment	Lagerborg et al. (2023)	ICE, real GDP, uncertainty, Real stock prices	1983Q3 – 2019Q4
TFP	Ben Zeev and Khan (2015)	Real GDP per capita, real stock prices per capita, labor productivity	1983Q3 – 2012Q1

Note: All specifications include lags of the dependent variable and the shock series as controls and are estimated with two lags. 'ICE' is the University of Michigan Index of Consumer Expectations. Uncertainty is measured as the 12-month Jurado et al. (2015) uncertainty index.

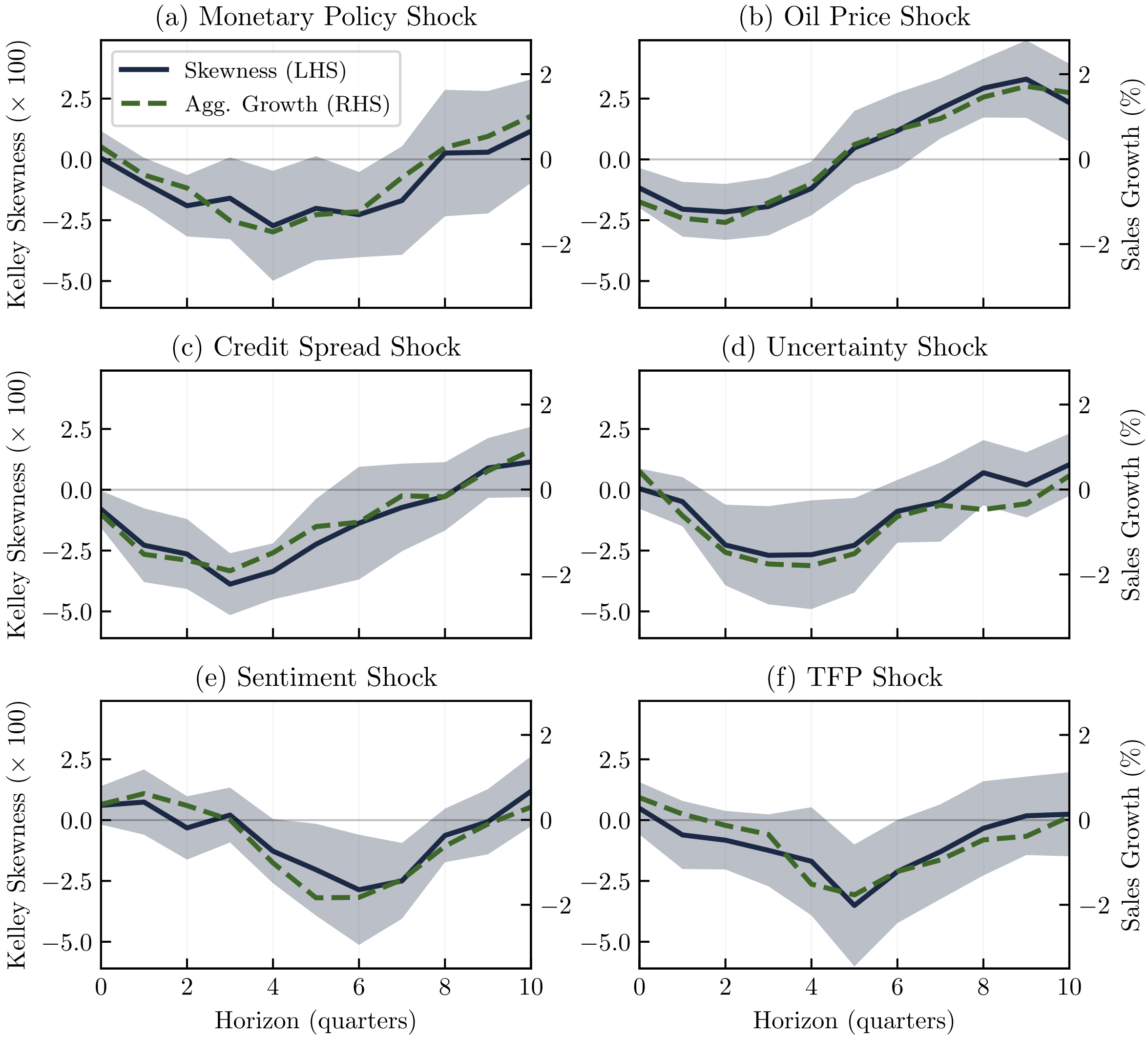
large and small firms. Table 2 summarizes the specifications, and Appendix D.5 provides details on the shock series and robustness checks.

Aggregate evidence. Figure 9 reports the impulse responses of cross-sectional skewness (dark blue solid line, left axis) and aggregate sales growth (green, dashed line, right axis) to the six identified aggregate shocks. A clear pattern emerges across all specifications: contractionary shocks lower cross-sectional skewness, and aggregate sales growth moves closely alongside it. Following an adverse one-standard-deviation shock, the skewness index falls by about 0.02 to 0.06 points, with the response typically peaking after 4 to 6 quarters and fading within about 10 quarters. The decline is largest for credit shocks and smallest for oil shocks. Across shock series, the correlation between the impulse responses of growth and skewness ranges from 0.89 to 0.98. Aggregate shocks therefore appear capable of (1) inducing significant movements in skewness and (2) generating strong comovement between sales growth and skewness, in line with our theoretical predictions.

Note that these results closely mirror the reduced-form evidence in Figure 2. Cross-sectional skewness moves closely with aggregate growth across many US recessions (including the Covid recession), suggesting that procyclical skewness is a robust business cycle fact not specific to particular types of recessions. It is therefore encouraging that different types of shocks — all considered potentially important drivers of the US business cycle — induce this pattern.

Heterogeneity by firm size. We now turn to the second empirical prediction, that the skewness effect of aggregate shocks should be stronger among larger firms, which operate with greater market power. To test this implication, we split the local projections in Equation (33) by firm size and compute skewness indices for large and small firms separately. Figure 10 shows a clear shows a clear size gradient of skewness. Across all shocks, the skewness index of large-firm growth rates (dark blue solid lines) declines

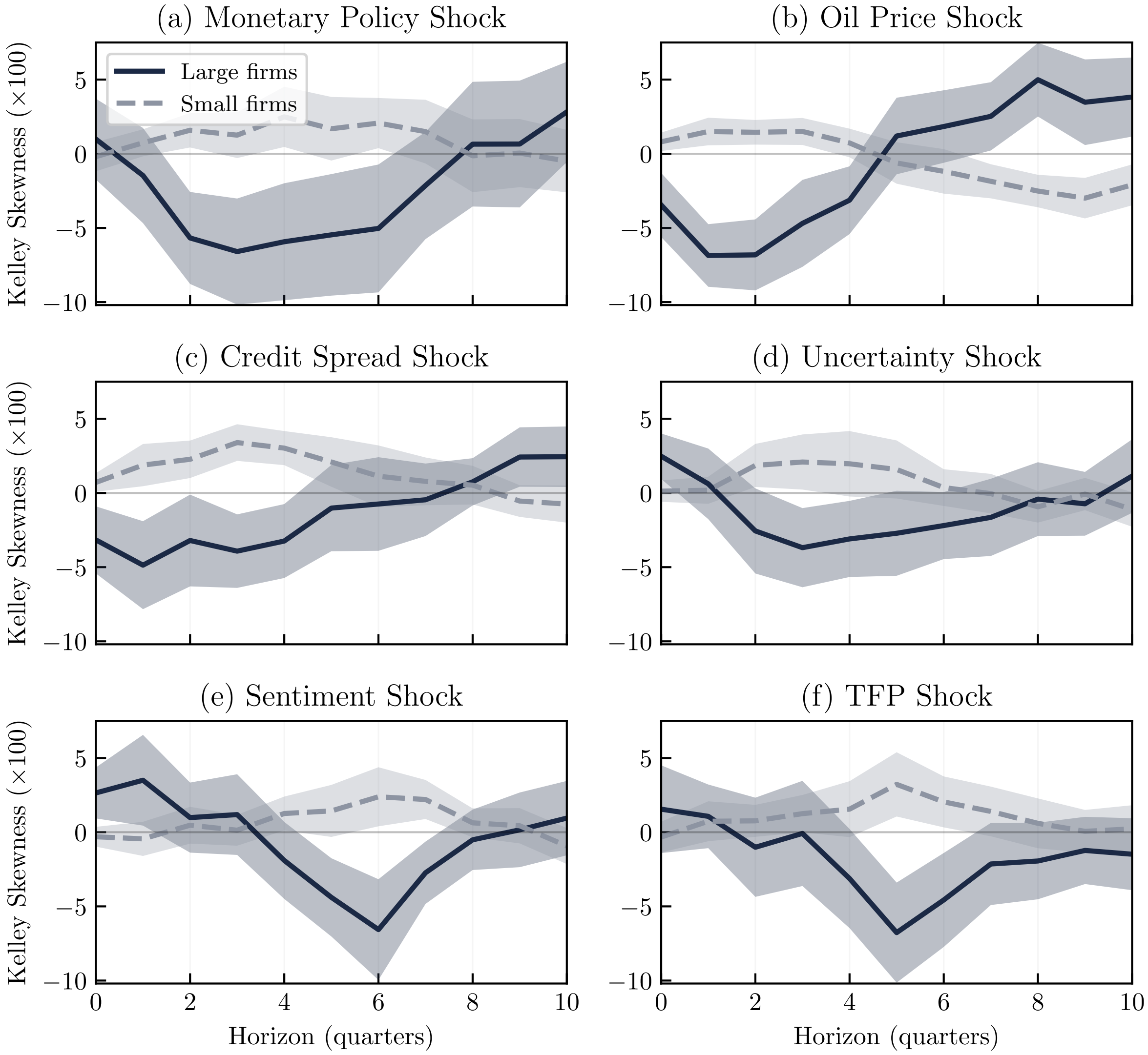
Figure 9: Comovement of growth and skew after aggregate shocks



Note: The 90% confidence bands are based on Newey-West standard errors. Shock magnitudes are normalized to be one standard deviation. The signs of the sentiment and the TFP shock are reversed to be contractionary.

by about 0.04 to 0.06 points, while the corresponding effect for small firms (dashed gray lines) remains close to zero. This pattern maps directly into the mechanism of the model. Firms with greater market power have more concave output policies, and greater concavity implies a stronger skewness reaction to a common aggregate disturbance. Since large firms operate with higher market power, their growth-rate skewness responds much more strongly to aggregate shocks. In Appendix D.4, we show that the same size pattern emerges from a bottom-up approach to impulse responses, confirming the robustness of these findings.

Figure 10: Skewness response of small and large firms



Note: The 90% confidence bands are based on Newey-West standard errors. Shock magnitudes are normalized to be one standard deviation. The signs of the sentiment and the TFP shock are reversed to be contractionary.

6 Conclusion

This paper has documented new evidence on the shape of firm growth distributions over the business cycle. Using Compustat data, we confirmed that skewness is procyclical, becoming negative in recessions and positive in booms. We then showed that this pattern is strongly size dependent: large firms display much larger swings in skewness than smaller firms. Finally, we established that countercyclical variance amplifies these effects, with increases in volatility leading to disproportionately negative skewness among the largest firms. Taken together, these findings point to a systematic size gradient of skewness in the cross-section of firms.

Since this size gradient cannot be explained by theories in which firms are fundamentally homogeneous, we set out to develop a simple framework to rationalize all empirical observations jointly. When firms possess market power, Marshall's Second Law of Demand implies that symmetric shocks translate into locally concave output adjustments. Along a path that starts at the price-taking benchmark, introducing market power makes the benchmark curvature $\rho^\star_\alpha$ more negative under transparent sufficient conditions. This mechanism generates systematically skewed growth responses, with the effect becoming stronger as market power rises locally away from competition and as the variance of shocks increases. The model rationalizes both the procyclicality of skewness and its dependence on firm size, and it matches the empirical impulse responses we estimate from the data. The cross-sectional evidence points in the same direction: skewness cyclicality rises across estimated-markup and profitability deciles and is greater in more concentrated industries. In this sense, skewness is not a primitive property of the shock distribution but an endogenous outcome of firm behavior under imperfect competition.

Appendix

Proofs

Proof of Proposition 1. The formulas follow by a second-order Taylor expansion of $Q^\star$ around the benchmark shock. The mean and variance use the first two terms of that expansion; the third central moment is the calculation in Lemma 2, applied to the period-t and period-$(t-1)$ shock components and then rescaled back to the notation of Proposition 1. ■

Lemma 2 (Small-scale third central moment and skewness)**.** *Let U_1, U_2, Z be independent real-valued random variables with continuous distributions such that:*

1. $U_1 \overset{d}{=} U_2$, $\mathbb{E}[U_i] = 0$, *and U_i is symmetric about* 0*;*
2. *Z is symmetric about 0 with* $\mathbb{E}[Z] = 0$*;*
3. $\mu_{4,U} := \mathbb{E}[U_1^4] < \infty$ *and* $\mu_{4,Z} := \mathbb{E}[Z^4] < \infty$.

Let $f \in C^2$ in a neighborhood of 0 and denote $f_1 := f'(0)$ and $f_2 := f''(0)$. Fix $a, b \in \mathbb{R}$, and define for small $\varepsilon > 0$

$$W_\varepsilon := f\big(\varepsilon(U_1 + Z)\big) - f\big(\varepsilon(bU_2 + aZ)\big).$$

Then,

$$\mu_3(\varepsilon) := \mathbb{E}\big[(W_\varepsilon - \mathbb{E}W_\varepsilon)^3\big] = \frac{3}{2} f_1^2 f_2\, \varepsilon^4 \operatorname{Cov}(D^2, H) + o(\varepsilon^4),$$
$$D := (U_1 - bU_2) + (1-a)Z, \qquad H := (U_1 + Z)^2 - (bU_2 + aZ)^2,$$
$$\operatorname{Cov}(D^2, H) = (1 - b^4)\operatorname{Var}(U_1^2) - 4(a-1)(ab^2+1)\,\sigma_U^2\sigma_Z^2 - (a^2-1)(a-1)^2 \operatorname{Var}(Z^2).$$

Moreover,

$$\mathrm{Var}(W_\varepsilon) = f_1^2\,\varepsilon^2\Big((1+b^2)\sigma_U^2 + (a-1)^2\sigma_Z^2\Big) + o(\varepsilon^2),$$

and hence the skewness coefficient is

$$\gamma_1(\varepsilon) := \frac{\mu_3(\varepsilon)}{\mathrm{Var}(W_\varepsilon)^{3/2}} = \varepsilon \cdot \frac{f_2}{|f_1|} \cdot \frac{\frac{3}{2}\,\mathrm{Cov}(D^2, H)}{\Big(\big((1+b^2)\sigma_U^2 + (a-1)^2\sigma_Z^2\big)\Big)^{3/2}} + o(\varepsilon).$$

Proof. Set $X_1 := U_1+Z$, $X_2 := bU_2+aZ$, $D := X_1-X_2$, and $H := X_1^2-X_2^2$. Taylor-expand f at 0 to write $W_\varepsilon = f_1\varepsilon D+\frac{f_2}{2}\varepsilon^2 H+o(\varepsilon^2)$. Since D is symmetric about 0, all odd moments of D vanish. Expanding the cube of the centered W_ε, the ε^3 term drops out by this symmetry and the leading contribution to μ_3 is $\frac{3}{2}f_1^2 f_2\,\varepsilon^4\,\mathrm{Cov}(D^2, H) + o(\varepsilon^4)$. Expanding D^2H and taking expectations term-by-term using independence, zero means, and $U_1 \overset{d}{=} U_2$ yields the stated formula for $\mathrm{Cov}(D^2, H)$. The variance follows from $\mathrm{Var}(W_\varepsilon) = f_1^2\varepsilon^2\mathrm{Var}(D) + o(\varepsilon^2)$. ■

Proof. Set $X_1 := U_1 + Z$, $X_2 := bU_2 + aZ$, $D := X_1 - X_2$, and $H := X_1^2 - X_2^2$. Taylor expansion gives

$$W_\varepsilon = f_1\varepsilon D + \frac{f_2}{2}\varepsilon^2 H + o(\varepsilon^2).$$

The random variable $D = (U_1 - bU_2) + (1-a)Z$ is symmetric, so $\mathbb{E}[D] = \mathbb{E}[D^3] = 0$. With $H_0 := H - \mathbb{E}[H]$,

$$W_\varepsilon - \mathbb{E}[W_\varepsilon] = f_1\varepsilon D + \frac{f_2}{2}\varepsilon^2 H_0 + o(\varepsilon^2),$$

and cubing gives

$$\mu_3(\varepsilon) = \frac{3}{2}f_1^2 f_2\varepsilon^4\,\mathrm{Cov}(D^2, H) + o(\varepsilon^4).$$

A direct expansion of D^2H, using independence, zero means, and $U_1 \overset{d}{=} U_2$, yields

$$\mathrm{Cov}(D^2, H) = (1-b^4)\mathrm{Var}(U_1^2) - 4(a-1)(ab^2+1)\sigma_U^2\sigma_Z^2 - (a^2-1)(a-1)^2\mathrm{Var}(Z^2).$$

Finally, $\mathrm{Var}(W_\varepsilon) = f_1^2\varepsilon^2\mathrm{Var}(D) + o(\varepsilon^2)$ with $\mathrm{Var}(D) = (1+b^2)\sigma_U^2 + (a-1)^2\sigma_Z^2$. Combining these expressions gives the stated standardized skewness expansion. ■

Proof of Lemma 1. Differentiating $\pi(q;\epsilon) = qp(q) - e^{\epsilon+C}q^\eta$ gives $p(q) + qp'(q) = \eta e^{\epsilon+C}q^{\eta-1}$, which is (14). The right-hand side is strictly increasing in q because $\eta > 1$.

It remains to show that marginal revenue is strictly decreasing on the relevant domain. Since $\mathrm{mr}(q) = p(q) + qp'(q)$, $\mathrm{mr}'(q) = 2p'(q) + qp''(q)$. Log-concavity of p implies $p''(q)p(q) - p'(q)^2 \le 0$, or $p''(q)/p'(q) \ge p'(q)/p(q)$ because $p'(q) < 0$. Therefore

$$\begin{aligned}\mathrm{mr}'(q) &= p'(q)\left(2 + q\frac{p''(q)}{p'(q)}\right)\\ &\le p'(q)\left(2 + q\frac{p'(q)}{p(q)}\right) = p'(q)\big(2 + \mathcal{E}p(q)\big).\end{aligned}$$

Since $p'(q) < 0$ and $\mathcal{E}p(q) \in (-1, 0)$ whenever marginal revenue is positive, the right-hand side is strictly negative. Hence mr is strictly decreasing wherever the firm operates.

By Assumption 1, marginal revenue is positive at some finite output level, while marginal cost is continuous, strictly increasing, and tends to zero as $q \downarrow 0$. Hence the two sides of (14) cross exactly once.

Taking logs of (14) gives (15). Since the left-hand side is strictly decreasing in $\widehat{q} = \log q$, its local inverse $Q^\star$ is well defined and strictly decreasing in ϵ. Finally, positivity of marginal revenue implies $1 + \mathcal{E}p(q) > 0$, and since $p'(q) < 0$ we obtain $\mathcal{E}p(q) \in (-1, 0)$. The markup formula follows immediately. ■

Lemma 3 (Local Curvature of Output)**.** *Let $\widehat{q}^\star := Q^\star(\epsilon^\star)$ and $q^\star := e^{\widehat{q}^\star}$. The policy $Q^\star$ is the local inverse of H around $\widehat{q}^\star$. If $H'(\widehat{q}^\star) \neq 0$, then*

$$Q^{\star\prime}(\epsilon^\star) = \frac{1}{H'(\widehat{q}^\star)}, \qquad Q^{\star\prime\prime}(\epsilon^\star) = -\frac{H''(\widehat{q}^\star)}{H'(\widehat{q}^\star)^3}, \qquad \rho^\star = \frac{Q^{\star\prime\prime}(\epsilon^\star)}{|Q^{\star\prime}(\epsilon^\star)|} = \frac{H''(\widehat{q}^\star)}{H'(\widehat{q}^\star)^2}. \tag{34}$$

Moreover,

$$H'(\widehat{q}^\star) = \mathcal{E}\mathrm{mr}(q^\star) - (\eta - 1), \qquad H''(\widehat{q}^\star) = \left.\frac{d\,\mathcal{E}\mathrm{mr}(q)}{d\log q}\right|_{q=q^\star}. \tag{35}$$

Hence local MSLD′ at $q^\star$ is equivalent to $Q^{\star\prime\prime}(\epsilon^\star) < 0$, and therefore to $\rho^\star < 0$.

Proof. The first-order condition is $H(\widehat{q}) = \epsilon$, with $H(\widehat{q}^\star) = \epsilon^\star$. By the inverse-function theorem, $Q^\star = H^{-1}$ locally. Differentiating $H(Q^\star(\epsilon)) = \epsilon$ once and twice gives

$$H'(Q^\star(\epsilon))Q^{\star\prime}(\epsilon) = 1, \qquad H''(Q^\star(\epsilon))(Q^{\star\prime}(\epsilon))^2 + H'(Q^\star(\epsilon))Q^{\star\prime\prime}(\epsilon) = 0.$$

Evaluating at $\epsilon^\star$ yields the first two identities in (34). Since $H'(\widehat{q}^\star) < 0$, $|Q^{\star\prime}(\epsilon^\star)| = -1/H'(\widehat{q}^\star)$, giving the curvature identity.

Differentiating the definition of H gives $H'(\widehat{q}) = d\log \mathrm{mr}(e^{\widehat{q}})/d\widehat{q} - (\eta - 1) = \mathcal{E}\mathrm{mr}(e^{\widehat{q}}) - (\eta - 1)$ and $H''(\widehat{q}) = d\,\mathcal{E}\mathrm{mr}(q)/d\log q|_{q=e^{\widehat{q}}}$, which gives (35).

Finally, $\mathcal{E}\mathrm{mr}(q^\star) < 0$ and $\eta > 1$ imply $H'(\widehat{q}^\star) < 0$, so $Q^{\star\prime\prime}(\epsilon^\star)$ has the same sign as $H''(\widehat{q}^\star)$. Local MSLD′ means $d|\mathcal{E}\mathrm{mr}(q)|/dq|_{q=q^\star} > 0$, equivalently $d\,\mathcal{E}\mathrm{mr}(q)/dq|_{q=q^\star} < 0$ because $\mathcal{E}\mathrm{mr}(q) < 0$. By (35), this is equivalent to $H''(\widehat{q}^\star) < 0$, and the sign equivalences for $Q^{\star\prime\prime}(\epsilon^\star)$ and $\rho^\star$ follow. ■

Proof of Proposition 2. Step 1. By local MSLD, $\varepsilon^\star < 0$ and $\psi^\star > 0$. Weak local ISID means $\Pi^\star \geq 0$. Evaluating (42) at the reference quantity $q = q^\star$ gives

$$\left.\frac{dm(q)}{d\log q}\right|_{q=q^\star} = \frac{\varepsilon^\star}{\left(1+\varepsilon^\star\right)^2}\Big[(1+\varepsilon^\star)\Pi^\star + \psi^\star\left(1+\varepsilon^\star\right)^2 + (\psi^\star)^2\Big].$$

Because $\varepsilon^\star \in (-1, 0)$, the denominator is positive, the prefactor is negative, and the bracketed term is strictly positive. Hence $dm(q)/d\log q|_{q=q^\star} < 0$. Since $m(q) = \mathcal{E}\mathrm{mr}(q) < 0$, this is local MSLD′, and Lemma 3 implies $\rho^\star < 0$.

Step 2. Let $q(\bar c)$ solve $\max_{q\ge 0}\ p(q)q - \bar c q^\eta$, and let $p^{\mathrm{opt}}(\bar c) = p(q(\bar c))$. The first-order condition is

$$p(q) + qp'(q) = \eta \bar c\, q^{\eta-1}. \tag{36}$$

By the implicit function theorem,

$$\tau(\bar c) = \frac{dp^{\mathrm{opt}}(\bar c)}{d\bar c} = p'(q)\frac{dq}{d\bar c} = \frac{\eta\, p'(q)\, q^{\eta-1}}{2p'(q) + qp''(q) - \eta(\eta-1)\bar c\, q^{\eta-2}},$$

where $q = q(\bar c)$. Using (36), $qp'(q) = \varepsilon(q)p(q)$, and $\psi(q) = 1 - \varepsilon(q) + qp''(q)/p'(q)$ gives

$$\tau(q) = \frac{\eta\, \varepsilon(q)\, q^{\eta-1}}{\varepsilon(q)\big(\varepsilon(q) + \psi(q) + 2 - \eta\big) - (\eta-1)}. \tag{37}$$

Since $q(\bar c)$ is decreasing in $\bar c$, local IPT implies $\tau'(q) < 0$ along the optimum.

Write $D(q) = \varepsilon(q)(\varepsilon(q)+\psi(q)+2-\eta)-(\eta-1)$. Using $q\varepsilon'(q) = \varepsilon(q)\psi(q)$ and $q\psi'(q) = \Pi(q)$, differentiation yields

$$\frac{d\tau(q)}{dq} = \frac{\eta\, q^{\eta-2}\varepsilon(q)}{D(q)^2}\left\{\big(\psi(q)+\eta-1\big)D(q) - \varepsilon(q)\Big[\psi(q)\big(2\varepsilon(q)+\psi(q)+2-\eta\big) + \Pi(q)\Big]\right\}.$$

Because $D(q)^2 > 0$ and $\varepsilon(q) < 0$, the inequality $\tau'(q) < 0$ is equivalent to

$$\Pi(q) > \frac{\eta-1}{-\varepsilon(q)}\psi(q) + (\eta-1)^2\left(\frac{1}{-\varepsilon(q)} - 1\right) + (\eta-1)\big(1-\varepsilon(q)\psi(q)\big) + \psi(q)\big(-\varepsilon(q)\big). \tag{38}$$

Under local MSLD we have $\psi(q) \ge 0$ at the reference quantity, while $\eta > 1$ and $\varepsilon(q) \in (-1, 0)$ on the relevant domain. Hence every term on the right-hand side of (38) is nonnegative, and the last term is strictly positive when local MSLD is strict. Therefore $\Pi^\star > 0$, i.e. local ISID holds. ■

Proof of Corollary 1. For the price taker, $\log \bar p - \log \eta - (\eta-1)\widehat{q}_{\mathrm{pt}} - C = \epsilon$, so

$$Q^\star_{\mathrm{pt}}(\epsilon) = \frac{\log \bar p - \log \eta - C - \epsilon}{\eta - 1}.$$

This policy is affine, hence $Q^{\star\prime\prime}_{\mathrm{pt}}(\epsilon^\star) = 0$ and $\rho^\star_{\mathrm{pt}} = 0$. The price-setting statement follows from Lemma 3. ■

Proof of Corollary 2. By Proposition 2, local MSLD′ at the no-shock benchmark implies $\rho^\star < 0$. Under the maintained assumptions, Proposition 1 gives

$$\gamma_1(g_t) \simeq \rho^\star \cdot \frac{3}{2}\frac{(u_t^2 - u_{t-1}^2)(u_t - u_{t-1})^2\,\mathrm{Var}(\lambda^2)}{\big((u_t - u_{t-1})^2\,\mathrm{Var}(\lambda)\big)^{3/2}}.$$

The denominator is strictly positive and $(u_t - u_{t-1})^2 \ge 0$ for every t, so the sign is determined by

$$\rho^\star \cdot (u_t^2 - u_{t-1}^2).$$

On impact, $u_0 = 0$ and $u_1 = x > 0$, so $u_1^2 - u_0^2 > 0$. Since $\rho^\star < 0$, this implies $\gamma_1(g_1) < 0$.

For every later date, $u_t = \beta^{t-1}x$ with $\beta \in (0,1)$, so $u_t < u_{t-1}$ and therefore $u_t^2 - u_{t-1}^2 < 0$. The sign thus flips and $\gamma_1(g_t) > 0$ for all $t > 1$. The formula is linear in $\rho^\star$, so the amplitude is proportional to $|\rho^\star|$. ■

Proof of Proposition 3. Use the demand-shape coefficients of p from (18) and write $\varepsilon_0 := \varepsilon(q_0^\star)$, $\psi_0 := \psi(q_0^\star)$, and $\Pi_0 := \Pi(q_0^\star)$.

Step 1. For each α, let $\widehat{q}_\alpha^\star$ solve $H_\alpha(\widehat{q}_\alpha^\star) = \epsilon^\star$, and define $R(\alpha, \widehat{q}) := H_\alpha''(\widehat{q})/H_\alpha'(\widehat{q})^2$. Then $\rho_\alpha^\star = R(\alpha, \widehat{q}_\alpha^\star)$ by (27). Since $H_0(\widehat{q}) = \log \bar{p} - \log \eta - (\eta - 1)\widehat{q}$ is affine, $R(0, \widehat{q}) = 0$ for all $\widehat{q}$, so $d\rho_\alpha^\star/d\alpha|_{0^+} = \partial_\alpha R(0, \widehat{q}_0^\star)$.

From (26),

$$H_\alpha'(\widehat{q}) = -(\eta - 1) + \alpha\varepsilon(q) + \frac{\alpha\varepsilon(q)\psi(q)}{1 + \alpha\varepsilon(q)}, \qquad q = e^{\widehat{q}}. \tag{39}$$

At $\alpha = 0$, this gives $H_0'(\widehat{q}_0^\star) = -(\eta - 1)$.

Apply the operator $D = q\frac{d}{dq}$ to (39). Because $D\varepsilon = \varepsilon\psi$ and $D\psi = \Pi$, we obtain

$$\partial_\alpha H_\alpha''(\widehat{q})|_{\alpha=0} = \varepsilon(q)\big(\psi(q) + \psi(q)^2 + \Pi(q)\big). \tag{40}$$

Since $H_0''(\widehat{q}_0^\star) = 0$, differentiating $R(\alpha, \widehat{q}) = H_\alpha''(\widehat{q})/H_\alpha'(\widehat{q})^2$ at $\alpha = 0$ gives

$$\partial_\alpha R(0, \widehat{q}_0^\star) = \frac{\partial_\alpha H_\alpha''(\widehat{q}_0^\star)|_{\alpha=0}}{H_0'(\widehat{q}_0^\star)^2}.$$

Substituting (40) and $H_0'(\widehat{q}_0^\star) = -(\eta - 1)$ gives

$$\left.\frac{d\rho_\alpha^\star}{d\alpha}\right|_{0^+} = \frac{\varepsilon_0}{(\eta - 1)^2}\big(\psi_0 + \psi_0^2 + \Pi_0\big),$$

which is (28).

Step 2. If local MSLD holds at $q_0^\star$, then $\psi_0 > 0$. If weak local ISID holds, then $\Pi_0 \geq 0$. Since $\varepsilon_0 < 0$, the formula above immediately implies (29). If instead local IPT and local MSLD hold for p at $q_0^\star$, then the implication (21) from Proposition 2, evaluated under p, gives local ISID for p at $q_0^\star$. The previous argument then yields the final sentence of the proposition. ■

Proof of Corollary 3. For any $\widetilde{\alpha} > 0$ on the path, the coefficient identities $\varepsilon_{\widetilde{\alpha}} = \widetilde{\alpha}\varepsilon$, $\psi_{\widetilde{\alpha}} = \psi$, and $\Pi_{\widetilde{\alpha}} = \Pi$ imply that local MSLD and weak local ISID for $p_{\widetilde{\alpha}}$ at $q_0^\star$ are equivalent to the same restrictions for p. By Proposition 3, $d\rho_\alpha^\star/d\alpha|_{0^+} < 0$. The function $\alpha \mapsto \rho_\alpha^\star$ is continuous because H_α is smooth in both arguments and $\widehat{q}_\alpha^\star$ is locally smooth by the implicit function theorem. Hence there exists $\bar{\alpha} > 0$ such that $\rho_\alpha^\star$ is negative and strictly decreasing on $(0, \bar{\alpha})$.

For a fixed shock process, Proposition 1 shows that moment-based skewness is proportional to $\rho_\alpha^\star$, up to the common shock term. Therefore the sign pattern of skewness is the same for every α with $\rho_\alpha^\star < 0$, and the absolute amplitude is strictly larger for the larger-α firm whenever the shock term is nonzero. The same local curvature ranking carries over to Kelley skewness by the local bridge in Section 3. ∎

Additional Theory Results

Lemma 4 (Local Elasticity and Marginal-Revenue Formulas)**.** *Let $m(q) := \mathcal{E}\mathrm{mr}(q)$. Then*

$$m(q) = \varepsilon(q) + \frac{\varepsilon(q)}{1+\varepsilon(q)}\psi(q), \tag{41}$$

$$\frac{dm(q)}{d\log q} = \frac{\varepsilon(q)}{\big(1+\varepsilon(q)\big)^2}\Big[\big(1+\varepsilon(q)\big)\Pi(q) + \psi(q)\big(1+\varepsilon(q)\big)^2 + \psi(q)^2\Big]. \tag{42}$$

Proof. Write $\mathrm{mr}(q) = p(q)\big(1+\varepsilon(q)\big)$ with $\varepsilon(q) = \mathcal{E}p(q)$. Taking elasticities gives $m(q) = \mathcal{E}\mathrm{mr}(q) = \mathcal{E}p(q) + \mathcal{E}(1+\varepsilon(q)) = \varepsilon(q) + \varepsilon(q)\psi(q)/(1+\varepsilon(q))$, which is (41). Since $d\varepsilon(q)/d\log q = \varepsilon(q)\psi(q)$ and $d\psi(q)/d\log q = \Pi(q)$, differentiating (41) yields

$$\frac{dm(q)}{d\log q} = \frac{\varepsilon(q)}{\big(1+\varepsilon(q)\big)^2}\Big[\big(1+\varepsilon(q)\big)\Pi(q) + \psi(q)\big(1+\varepsilon(q)\big)^2 + \psi(q)^2\Big],$$

which is (42). ∎

Lemma 5 (Why the benchmark must be constant)**.** *Consider the generalized family*

$$\widetilde{p}_\alpha(q) = p(q)^\alpha \bar{p}(q)^{1-\alpha}, \qquad \alpha \in [0,1], \tag{43}$$

where both p and $\bar{p}$ are strictly positive and differentiable. Let

$$\varepsilon(q) := \frac{qp'(q)}{p(q)}, \qquad \bar{\varepsilon}(q) := \frac{q\bar{p}'(q)}{\bar{p}(q)}. \tag{44}$$

Then the induced markup is

$$\mu_\alpha(q) = \frac{1}{1 + \alpha\varepsilon(q) + (1-\alpha)\bar{\varepsilon}(q)}. \tag{45}$$

Hence $\mu_0(q) = 1$ for all q if and only if $\bar{\varepsilon}(q) = 0$ for all q, i.e. if and only if $\bar{p}(q)$ is constant. Moreover,

$$\frac{\partial \mu_\alpha(q)}{\partial \alpha} = -\frac{\varepsilon(q) - \bar{\varepsilon}(q)}{\big(1 + \alpha\varepsilon(q) + (1-\alpha)\bar{\varepsilon}(q)\big)^2}, \tag{46}$$

so a nonconstant benchmark need not make markups increase with α.

Proof. Since $\log \widetilde{p}_\alpha(q) = \alpha \log p(q) + (1-\alpha)\log \bar{p}(q)$, differentiating with respect to $\log q$ gives $q\widetilde{p}'_\alpha(q)/\widetilde{p}_\alpha(q) = \alpha\varepsilon(q) + (1-\alpha)\bar{\varepsilon}(q)$. Thus $\mu_\alpha(q) = 1/[1+\alpha\varepsilon(q)+(1-\alpha)\bar{\varepsilon}(q)]$. At $\alpha = 0$, $\mu_0(q) = 1/(1+\bar{\varepsilon}(q))$, so $\mu_0(q) = 1$ for all q if and only if $\bar{\varepsilon}(q) = q\bar{p}'(q)/\bar{p}(q) = 0$ for all q, i.e. if and only if $\bar{p}$ is constant.

Differentiating $\mu_\alpha(q)$ with respect to α yields

$$\frac{\partial \mu_\alpha(q)}{\partial \alpha} = -\frac{\varepsilon(q) - \bar{\varepsilon}(q)}{\left(1+\alpha\varepsilon(q)+(1-\alpha)\bar{\varepsilon}(q)\right)^2},$$

which shows that the sign of the markup response is not pinned down by α unless the benchmark is constant or an additional restriction is imposed. ∎

Proposition 4 (CES as a knife-edge benchmark)**.** *If*

$$p(q) = Aq^{-\chi}, \qquad A > 0, \qquad \chi \in (0,1), \tag{47}$$

then $\varepsilon(q) \equiv -\chi$*,* $\psi(q) \equiv 0$*, and* $\Pi(q) \equiv 0$*. Consequently,*

$$\rho^\star_\alpha = 0 \qquad \forall \alpha \in [0,1]. \tag{48}$$

CES therefore remains the knife-edge case in which the policy is locally linear and skewness is absent to second order.

Proof. If $p(q) = Aq^{-\chi}$, then $\varepsilon(q) = qp'(q)/p(q) = -\chi$ is constant, so $\psi(q) \equiv 0$ and $\Pi(q) \equiv 0$. Substituting into the first-order condition gives

$$H_\alpha(\widehat{q}) = (1-\alpha)\log \bar{p} + \alpha \log A + \log(1-\alpha\chi) - \log \eta - (\eta - 1 + \alpha\chi)\widehat{q},$$

which is affine in $\widehat{q}$ for every α. Therefore $H''_\alpha(\widehat{q}) = 0$ and thus $\rho^\star_\alpha = 0$ for all $\alpha \in [0,1]$. ∎

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Supplemental Online Appendix

Monopolistically Skewed Business Cycles

A Supply Side Skewness

Up until now, we have focused on (negative) skew driven by the shape of the demand curve, neglecting contributions of the cost function to skewness. We discuss two types of cost functions which can contribute to skewed growth rates. Either approach rests on the assumption that adjustment of some status quo (or the 'current state') is costly in one direction and cheap in the other. This leads to a cost function with a kink located at previous-period output, which immediately implies local log-convexity of the cost function and locally left-skewed values of $\widehat{q}$.

We start from a price taking firm's static profit maximization problem (suppressing price taker subscripts),

$$\max_q q\,[\bar{p} - c(q)], \qquad c(q) = e^{\epsilon}\Psi(q).$$

Here, $\bar{p}$ is the given output price, $\Psi(q)$ is the baseline cost function, and the exponential term e^{ϵ} is a multiplicative cost shifter. The first-order condition equates marginal revenue and marginal cost. Differentiating, we obtain

$$0 = \bar{p} - e^{\epsilon}\big(\Psi(q) + q\Psi'(q)\big) \equiv \bar{p} - e^{\epsilon}\mathfrak{mc}(q),$$

where $\mathfrak{mc}(q)$ is the slope of total cost. Taking logs yields $\ln \mathfrak{mc}(q) = \log \bar{p} - \epsilon$. This condition allows us to study how equilibrium output responds to the cost shifter. Differentiating the FOC, one can show that

$$\frac{d\widehat{q}}{d\epsilon} = -\frac{1}{\mathcal{E}\mathfrak{mc}(q)}, \qquad \frac{d^2\widehat{q}}{d\epsilon^2} = \frac{1}{(\mathcal{E}\mathfrak{mc}(q))^2}\,(\mathcal{E}\mathfrak{mc})'(q)\,\frac{dq}{d\epsilon}.$$

Since $dq/d\epsilon < 0$ (higher costs reduce output), concavity of $\widehat{q}$ in ϵ requires that $(\mathcal{E}\mathfrak{mc})'(q) \geq 0$, i.e. that the elasticity of marginal cost is increasing in quantity. Put differently: if $\mathfrak{mc}(q)$ is log-convex, then $\widehat{q}$ is locally concave in the cost shifter ϵ. Now consider how different frictions shape $\Psi(q)$, and thereby $\mathfrak{mc}(q)$.

Capacity adjustments. If producing beyond installed capacity $\bar{q}$ incurs a convex penalty on top of baseline costs, marginal cost jumps at $\bar{q}$ and rises steeply beyond it, making $\mathcal{E}\mathfrak{mc}(q)$ increasing and thus $\widehat{q}$ locally concave in ϵ. Expansions beyond capacity are disproportionately costly, so output falls quickly in response to adverse cost shocks but rises only sluggishly. Empirical support for such asymmetric adjustment costs comes from the investment literature (Cooper and Haltiwanger, 2006), and their inclusion in quantitative macro models is standard since Christiano et al. (2005).

Customer acquisition. If sales depend on a pre-existing customer base b plus new customers acquired at convex cost, then expanding output beyond b requires increasingly expensive acquisition effort, while contracting merely lets the base erode. This again makes $\mathcal{E}\mathfrak{mc}(q)$ increasing beyond b, ensuring local concavity of $\widehat{q}$. Models embedding customer bases into the firm problem include Gourio and Rudanko (2014), Roldan-Blanco and Gilbukh (2021), and Ignaszak and Sedlácek (2025).

In both cases, the critical feature is that the marginal cost function becomes increasingly steep as output rises. This makes the mapping from cost shocks ϵ to equilibrium quantities concave in logs, and thereby generates the negative skewness in firm-level growth rates. Existence of log-convexities in firms' cost functions can explain some pattern in Figure 3 (b). For a small size cutoff quantile, the plotted regression coefficients taper off and become constant at about 2.2. This suggests that growth rate skewness is still procyclical, even if market power, α, becomes small. Part of the reason may be that shocks are inherently skewed, but a more structural explanation is existence of log-convex cost functions even for price takers.

Yet, cost functions are unable to generate size dependence of skewness. To this observation, a demand-side oriented explanation caters nicely. Second, the approaches discussed above—capacity adjustments and customer acquisition cost—each only guarantee local skewness: growth rates relative to the steady state are skewed locally around long run output, q_0. That means, if the shock is not mean zero but, say, has a positive support such that *every* firm i in the cross-section adjusts to some $q_i < q_0$, then the negative skewness in growth rates is no longer guaranteed. In contrast, invoking ISID as a demand curve property immediately guarantees global left-skewness of growth rates, $\ln q/q_0$.

B Homothetic Demand Industry Equilibrium

Consider a representative consumer with fixed nominal expenditure $E > 0$ and homothetic preferences $U(\mathbf{q}) = F\big(h(\mathbf{q})\big)$, where F is strictly increasing and h is homogeneous of degree one, over a continuum of differentiated varieties supplied under monopolistic competition. The expenditure function factorizes as $e(\mathbf{p}, \bar{h}) = P(\mathbf{p})\,\bar{h}$, where $P(\mathbf{p}) \equiv e(\mathbf{p}, 1)$ is the unit-cost function (homogeneous of degree one in $\mathbf{p}$). By Shephard's lemma the Hicksian demand is $h_i(\mathbf{p}, \bar{h}) = \bar{h}\,\partial P/\partial p_i$; substituting $\bar{h} = E/P$ gives Marshallian demand

$$q_i(\mathbf{p}, E) = \frac{E}{P(\mathbf{p})}\,\frac{\partial P(\mathbf{p})}{\partial p_i} = \frac{E}{P(\mathbf{p})}\,D_i\!\left(\frac{\mathbf{p}}{P(\mathbf{p})}\right),$$

where $D_i(\mathbf{r}) \equiv \partial P(\mathbf{r})/\partial r_i$ depends on the full normalized price vector $\mathbf{r} = \mathbf{p}/P(\mathbf{p})$ (using homogeneity of degree zero of $\partial P/\partial p_i$).

Firms have isoelastic variable costs $C_i(q) = e^{u(\xi+\lambda_i)}q^{\eta}$ with $\eta > 1$, where u is an aggregate shock, ξ is common exposure, and λ_i is idiosyncratic exposure. There is a measure-zero set of "heterogeneous firms" (type A, $\lambda_i \neq 0$) and a unit mass of "fringe" firms (type B, $\lambda_i = 0$). Since type-A firms are measure zero they do not affect the price index or any aggregate object pinned down by the fringe.

Proposition A.5 (Common-scaling separability under homotheticity)**.** *Suppose (i) preferences are homothetic with Marshallian demands*

$$q_i(\mathbf{p}, E) = \frac{E}{P(\mathbf{p})}\,D_i\!\left(\frac{\mathbf{p}}{P(\mathbf{p})}\right),$$

(ii) heterogeneous firms (type A) have measure zero, while the fringe (type B) has unit mass and $\lambda = 0$, and (iii) for each u there exists a symmetric interior equilibrium for fringe firms with well-defined perceived elasticity at the symmetric point. Then log output of any heterogeneous firm i admits the decomposition

$$\log q_i(u, \lambda_i) = -\frac{\xi}{\eta}\,u + Q^{\star}(\lambda_i u),$$

where $Q^{\star}(\cdot)$ is independent of the common exposure ξ.

Proof. In the symmetric fringe equilibrium all fringe prices coincide at $p_B(u)$; since type-A firms have measure zero, $P(u) = \bar{P}\,p_B(u)$ for a constant $\bar{P} > 0$ (by HD1 of P). The normalized fringe price $r_B = p_B/P = 1/\bar{P}$ is independent of u. A fringe firm's profit at relative price r is $\pi_B = E\,r\,D(r) - e^{u\xi}(E/P(u))^{\eta}D(r)^{\eta}$. Revenue $E\,r\,D(r)$ is u-free, so for the FOC at r_B to hold for all u the cost prefactor $e^{u\xi}P(u)^{-\eta}$ must be constant, giving $P(u) = P(0)e^{\xi u/\eta}$ and $E/P(u) = (E/P(0))e^{-\xi u/\eta}$.

Firm i takes $P(u)$ and all other prices as given. Writing $r_i = p_i/P(u)$ and $\mathbf{r}_{-i} = \mathbf{p}_{-i}/P(u)$, revenue is $E\,r_i\,D_i(r_i; \mathbf{r}_{-i})$ (u-free) and cost simplifies to

$$e^{u(\xi+\lambda_i)}[(E/P(0))e^{-\xi u/\eta}D_i]^{\eta} = e^{u\lambda_i}(E/P(0))^{\eta}D_i^{\eta}.$$

All ξ-dependence cancels; the FOC w.r.t. r_i depends on u only through $e^{u\lambda_i}$. Every other type-A firm faces the same structure, so $\mathbf{r}^\star_{-i}$ depends on the respective $\lambda_j u$, not on ξu. Hence $\log q_i = -(\xi/\eta)\, u + Q^\star(\lambda_i u)$, where $Q^\star$ absorbs the equilibrium dependence on $r^\star_i$, $\mathbf{r}^\star_{-i}$, and D_i. ■

The key cancellation is $e^{u(\xi+\lambda_i)} \cdot (E/P(u))^\eta = e^{u\lambda_i}(E/P(0))^\eta$: the common cost factor $e^{u\xi}$ is absorbed exactly by $P(u)^\eta \propto e^{\xi u}$. The residual common scaling $e^{-\xi u/\eta}$ in output reflects the erosion of real expenditure as the price level rises with the common cost shock.

C Data Preparation

This appendix describes the construction of the firm-level panel used throughout the paper. The backbone of the analysis is quarterly data file from Compustat. In some exercises, we merge this sample with additional sources (CRSP, Worldscope Fundamentals, and I/B/E/S); these merges are described in separate subsections. Unless stated otherwise, the unit of observation is a firm-quarter, firms are identified by `gvkey`, and time is indexed by the calendar quarter reported in Compustat (`datacqtr`).

C.1 Compustat Quarterly Sample

Sample restrictions. We start from the full Compustat North America database at quarterly frequency. We restrict the sample to U.S. firms by requiring that the firm is both incorporated in the United States (`fic`=`USA`) and headquartered in the United States (`loc`=`USA`). We exclude non-operating and public establishments by dropping observations with SIC codes in $[9000, 9999]$. We drop firm-quarters with non-positive nominal sales and remove duplicate firm-quarter observations.

Real sales and growth rates. Nominal quarterly sales are deflated using the GDP price deflator (FRED series `GDPDEF`) to obtain real sales,

$$s_{i,t} \equiv \frac{\texttt{saleq}_{i,t}}{\texttt{GDPDEF}_t}.$$

and compute the year-on-year growth rate of real sales as

$$g_{i,t} \equiv \log s_{i,t} - \log s_{i,t-4}.$$

When real sales are missing in a quarter that is surrounded by non-missing observations for the same firm (i.e., observed at both $t-1$ and $t+1$), we fill the missing value by linear interpolation. If missing observations occur in adjacent quarters, we do not impute.

Aggregate and industry growth. We construct an aggregate real sales growth series by value-weighting firm-level growth rates using lagged real sales:

$$g_t \equiv \frac{\sum_i s_{i,t-4}\, g_{i,t}}{\sum_i s_{i,t-4}}. \tag{A.1}$$

This construction ensures that the aggregate series is based only on firms with observed sales in both t and $t-4$, and is therefore not mechanically affected by entry or exit. We analogously compute value-weighted sales growth at different levels of NAICS aggregation.

Firm characteristics. We construct several firm-level characteristics commonly used in the literature. Total debt is the sum of short-term and long-term debt (`dlcq`+`dlttq`), and total assets are `atq`. Leverage is defined as total debt divided by total assets. Net leverage subtracts net current assets from total debt before scaling by assets, where net current assets are current assets minus current liabilities (`actq-lctq`). Liquidity is measured as cash and short-term investments (`cheq`) divided by total assets. We additionally construct standard controls such as Tobin's Q, book-to-market, profitability proxies (EBITDA and cash flow), and sales-to-assets.

Cleaning and outlier treatment. We define the *full sample* as the set of firm-quarters that satisfy the geographic and industry restrictions above and for which year-on-year real sales growth can be computed. We then apply the following additional filters to obtain the *cleaned sample* on which most of our analysis is based, unless otherwise stated.

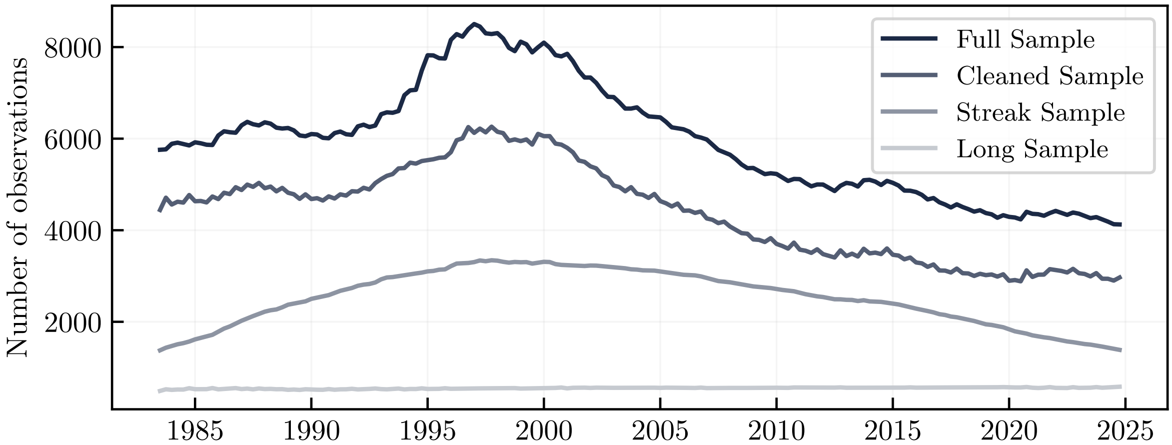
1. We drop firm-quarters with non-positive total assets or current assets. We also drop firm-quarters with non-positive cash and short-term investments (so that the liquidity ratio is well-defined).
2. We drop observations with extreme net current assets relative to total assets. Specifically, we keep only firm-quarters for which net current assets (current assets minus current liabilities) lie between -10 and $+10$ times total assets.
3. We drop observations with leverage below zero or above 10, where leverage is defined as total debt (short-term plus long-term) divided by total assets.
4. To remove extreme outliers in accounting levels, we drop firm-quarters in the top 0.1% of the sales-to-assets distribution.
5. We trim the top/bottom 1% of the cross-sectional distribution of year-on-year percent real sales growth within each quarter, and we drop observations with percent growth below -100%. We do not trim log growth

Sample period. Our baseline analysis focuses on the post-1983 period. We restrict the sample to 1983Q3–2024Q4.

Streak and long samples. Some parts of the analysis require substantial within-firm time-series variation. Following Ottonello and Winberry (2020), we construct a *streak sample* by retaining only sequences of at least 40 consecutive quarters for which year-on-year sales growth is observed for a given firm. In addition, to approximate a balanced panel, we define a *long sample* consisting of firms in the cleaned sample that are observed both near the beginning and near the end of the sample window.

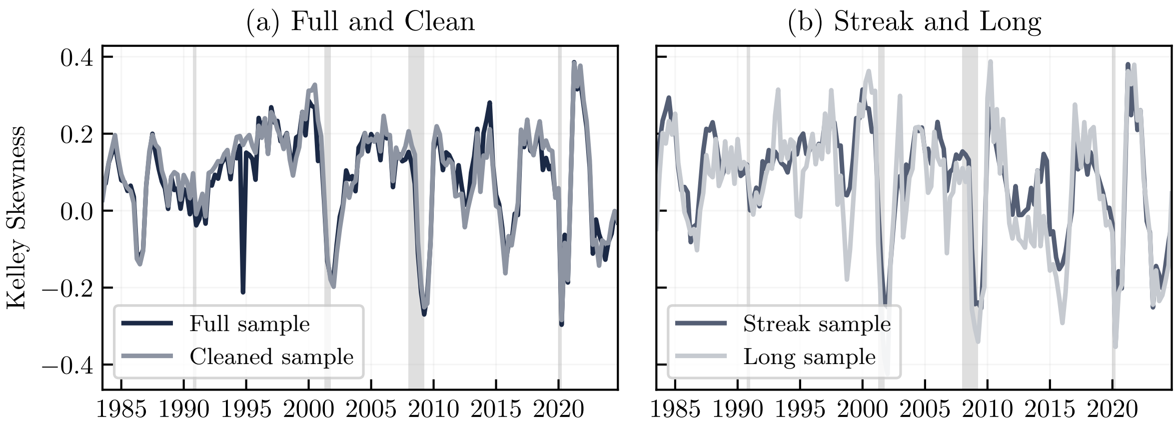
Figure A.1 shows the number of firm-level observations per quarter for the different samples. Despite differences in the number of observations, the skewness pattern across samples is very similar, see Figure A.2.

Figure A.1: Number of sales growth observations per quarter



Note: The full sample of growth rates covers 1962Q1-2022Q3. The other samples cover 1983Q3-2021Q4.

Figure A.2: Kelley skewness for different samples



Note: Skewness is computed using 90% Kelley skewness. The sample period is 1983Q3-2021Q4.

C.2 Compustat Global Quarterly Sample

Universe and currency conversion. We complement the North America sample with quarterly accounting data from Compustat Global. We drop non-operating and public establishments by excluding SIC codes in $[9000, 9999]$, remove duplicate firm-quarter observations, and drop firm-quarters with non-positive nominal sales. Because Compustat Global reports accounting items in local currencies, we first convert all key nominal balance-sheet and income-statement items to U.S. dollars using monthly exchange rates. Specifically, we match each firm-quarter to the average exchange rate over the calendar quarter (constructed from monthly averages by currency) and multiply the relevant flow and stock variables (e.g., sales, assets, debt, cash, costs) by this exchange rate. For observations already reported in U.S. dollars, the exchange rate is set to one.

Deflation, real sales, and growth rates. After converting nominal values into U.S. dollars, we deflate nominal sales using the GDP price deflator (FRED series `GDPDEF`) to obtain real sales. We compute year-on-year real sales growth in both logs and percent changes, and we treat isolated missing values in real sales by linear interpolation when a missing quarter is surrounded by non-missing observations for the same firm. As in the North America sample, we expand the panel to make missing firm-quarters explicit for the purposes of interpolation and streak construction.

Firm characteristics and cleaning. We construct the same set of firm characteristics as in the North America sample, including leverage, net current assets, liquidity, and standard profitability measures. We then apply the same outlier and plausibility filters: we drop firm-quarters with non-positive assets, current assets, or liquidity; drop observations with extreme net current assets relative to assets; drop observations with leverage outside a wide admissible range; trim the upper tail of sales-to-assets; and trim the top and bottom 1% of year-on-year percent real sales growth within each quarter while dropping percent growth below -100%.

Sample period and subsamples. The Global sample is restricted to the post-2000 period. We construct (i) a cleaned sample, (ii) a streak sample that retains firm-level sequences of at least 40 consecutive quarters with observed year-on-year percent sales growth, and (iii) a long sample intended to capture firms with observations both near the beginning and near the end of the sample window.

D Supplementary Empirical Results

D.1 Cyclicality regressions.

Table A.1 reports the simple time-series regressions discussed in Section 2. The dependent variable m_t is either the interquartile range s_t or Kelley skewness k_t of firm sales growth. The first specification relates each moment linearly to aggregate growth; the second lets the slope differ below and above average aggregate growth. The split specification confirms the geometry shown in Figure 2: dispersion rises on both sides of trend, while skewness is procyclical. The size panels confirm that the skewness-growth slope is substantially larger for large firms.

Table A.1: Cyclicality of Higher Moments of the Growth Rate Distribution

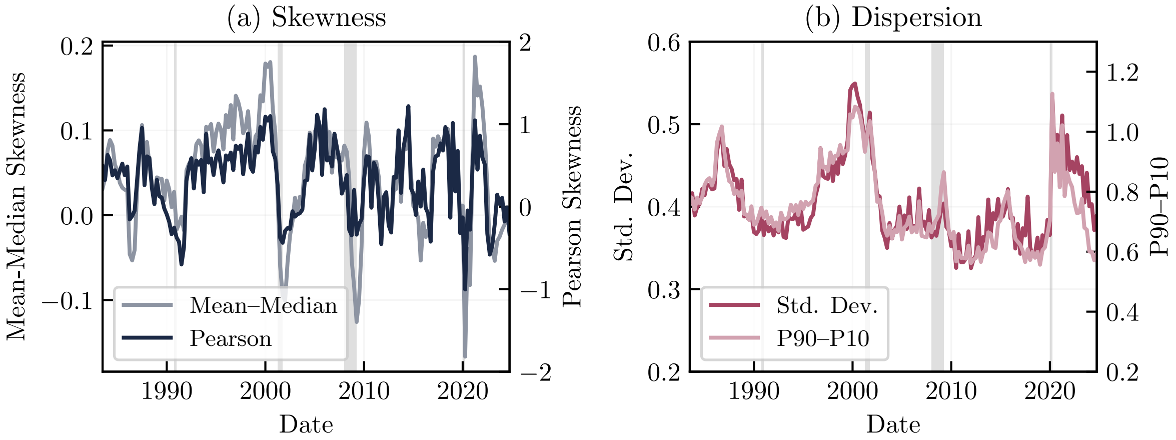
	s_t (IQR)		k_t (Skewness)	
Parameter	Estimate	SE	Estimate	SE
Panel A: Aggregate				
$m_t = \alpha + \beta\overline{g_t} + u_t$				
β	-0.057	(0.136)	1.743***	(0.088)
$m_t = \alpha + \beta^-\overline{g_t}\mathbf{1}\{\overline{g_t} < \overline{g}\} + \beta^+\overline{g_t}\mathbf{1}\{\overline{g_t} \geq \overline{g}\} + u_t$				
β^-	-0.763***	(0.122)	1.589***	(0.106)
β^+	0.262**	(0.103)	1.813***	(0.104)
Panel B: Small Firms (bottom 90% of time-t size distribution)				
$m_t = \alpha + \beta\overline{g_t} + u_t$				
β	-0.036	(0.133)	1.600***	(0.084)
$m_t = \alpha + \beta^-\overline{g_t}\mathbf{1}\{\overline{g_t} < \overline{g}\} + \beta^+\overline{g_t}\mathbf{1}\{\overline{g_t} \geq \overline{g}\} + u_t$				
β^-	-0.713***	(0.134)	1.456***	(0.096)
β^+	0.234**	(0.104)	1.657***	(0.101)
Panel C: Large Firms (top 10% of time-t size distribution)				
$m_t = \alpha + \beta\overline{g_t} + u_t$				
β	-0.233	(0.143)	3.104***	(0.303)
$m_t = \alpha + \beta^-\overline{g_t}\mathbf{1}\{\overline{g_t} < \overline{g}\} + \beta^+\overline{g_t}\mathbf{1}\{\overline{g_t} \geq \overline{g}\} + u_t$				
β^-	-0.879***	(0.071)	2.493***	(0.400)
β^+	0.516***	(0.133)	3.812***	(0.450)
Observations	162		162	

Note: Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.2 Robustness of Skewness Measures

Figure A.3 shows that the cyclical skewness pattern is not specific to Kelley skewness. Alternative quantile-based and moment-based skewness measures comove closely over the cycle, although Pearson skewness is more sensitive to extreme observations. The paper therefore uses Kelley skewness as the baseline because it is robust in short quarterly cross sections while preserving the same business-cycle pattern.

Figure A.3: Comparison of different skewness measures



Note. The first three measures are plotted on the left axis. Pearson skewness is plotted on the right axis.

D.3 Aggregate Factors Explain Skewness Fluctuations

This section asks whether the procyclical skewness in firm sales growth mainly reflects common business-cycle forces or skewed idiosyncratic shocks. In the absence of market-power curvature, growth is affine in aggregate factors and an idiosyncratic residual:

$$g_{i,t} = a + \widetilde{\boldsymbol{\lambda}}_i^T \boldsymbol{u}_t + w_{i,t}. \tag{A.2}$$

We estimate this decomposition with principal components, following Herskovic et al. (2016). PCA produces estimated common factors and firm loadings; for $K \in \{1, 4, 8\}$ retained components, we write the fitted common component and residual as

$$a_{i,t}^{(K)} = \sum_{k=1}^{K} \widehat{\lambda}_{i,k} \widehat{u}_{k,t}, \qquad w_{i,t}^{(K)} = g_{i,t} - a_{i,t}^{(K)}.$$

We then compute quarterly Kelley skewness for the aggregate component $\{a_{i,t}^{(K)}\}_i$ and the residual component $\{w_{i,t}^{(K)}\}_i$. To compare magnitudes, we decompose the Kelley-skewness numerator, $X_{0.9} - 2X_{0.5} + X_{0.1}$, into the aggregate and residual contributions evaluated at the growth-rate quantiles. The decomposition is approximate because firms need not have the same ranks in $g_{i,t}$, $a_{i,t}^{(K)}$, and $w_{i,t}^{(K)}$, but the approximation error is small in Figure A.4. This is a conservative test for our mechanism: if idiosyncratic skewness drives the skewness of firm growth, it should appear in the residual component.

Table A.2 shows that common factors account for most of the business-cycle variation in skewness. With one principal component, skewness in the aggregate component explains 79% of the variation in growth-rate skewness, while the idiosyncratic compo-

Table A.2: Common vs idiosyncratic drivers of skewness

No. Factors:	1	4	8
Correlations with skewness:			
$\rho\,(skew_u, skew_g)$	0.89	0.76	0.72
$\rho\,(skew_a, skew_g)$	0.68	0.72	0.81
Decomposition of variation in skewness:			
R_u^2	0.21	0.26	0.32
R_a^2	0.79	0.74	0.68
Fit of aggregate component:			
R_i^2 q(0.25)	0.01	0.09	0.16
R_i^2 q(0.5)	0.06	0.20	0.30
R_i^2 q(0.75)	0.18	0.36	0.47
Observations	398,316		

Note: Each column refers to a decomposition using a different number of principal components. The decomposition uses the weighted PCA algorithm of Delchambre (2015) with zero weights for missing values and unit weights for all other observations. The first two rows measure the correlation of 90% Kelley skewness in sales growth rates with the skewness in the idiosyncratic components ($skew_u$) or the aggregate components ($skew_a$). The following two rows decompose the variation in Kelley skewness into the contributions by skewness in the idiosyncratic part and skewness in the aggregate part. The last three rows show the 25, 50, and 75% quantile of the distribution across R^2 from firm-level time series regressions of the sales growth rate onto the aggregate component. The number of observations refers to the actual firm-quarter observations.

nent explains 21%. With eight factors, the aggregate share remains 68%. This does not happen because the common factors explain most firm-level growth variation: even with eight factors, the median firm-level R^2 is only 0.30. The result is therefore about higher moments, not about a mechanical high fit of firm growth.

Figure A.4 gives the same message visually. The aggregate component tracks the procyclical movement in growth skewness closely, while the idiosyncratic component adds little beyond that common-factor pattern. Thus skewness is not mainly a property of residual firm-specific shocks; it is largely embedded in the common component of firm growth.

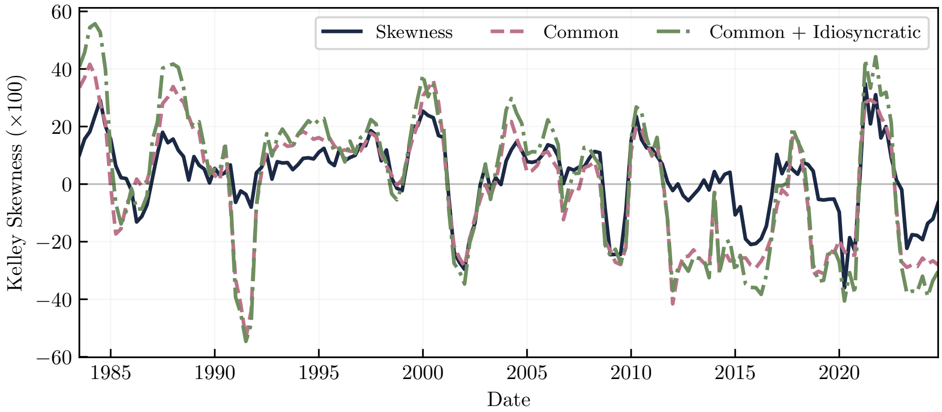
D.4 Robustness: Bottom-up impulse responses

The main text estimates impulse responses of aggregate growth and cross-sectional skewness directly. As a robustness check, we instead estimate firm-level responses to each aggregate shock and then reconstruct aggregate growth and skewness from the cross section of firm-level impulse responses. The firm-level local projection is

$$g_{i,t+h} = \alpha_{i,h} + \beta_{i,h}\text{shock}_t + \sum_{\ell=1}^{L} \gamma'_{i,\ell,h}\text{controls}_{i,t-\ell} + e_{i,t+h}, \tag{A.3}$$

where $\beta_{i,h}$ is firm i's response at horizon h. The controls mirror the aggregate specifications in Table A.4: two lags of the shock, dependent variable, GDP controls, and the

Figure A.4: Skewness in common vs idiosyncratic component



Note: The blue line is skewness in demeaned sales growth. The red line is the contribution of the aggregate component to the Kelley-skewness numerator. The green line adds the idiosyncratic contribution.

Table A.3: Firm-level local projections – Summary statistics

	Monetary	Oil	Credit	Uncertainty	Sentiment	TFP News
# Streaks	4,120	5,332	2,813	5,115	5,125	4,829
# Firms	4,017	5,061	2,813	4,893	4,902	4,651
Avg. # Obs.	64	70	84	66	66	63
Avg. R^2	0.33	0.29	0.25	0.26	0.32	0.31
Sign. IRFs (%)	77	80	83	78	82	80
$Q_{0.1}^{\text{IRF}}$ (%)	-5.4	-14.3	-2.1	-5.0	-4.5	-5.4
$Q_{0.5}^{\text{IRF}}$ (%)	-0.16	-0.14	-0.33	-0.30	-0.33	-0.39
$Q_{0.9}^{\text{IRF}}$ (%)	5.0	13.5	1.2	4.1	3.8	4.7

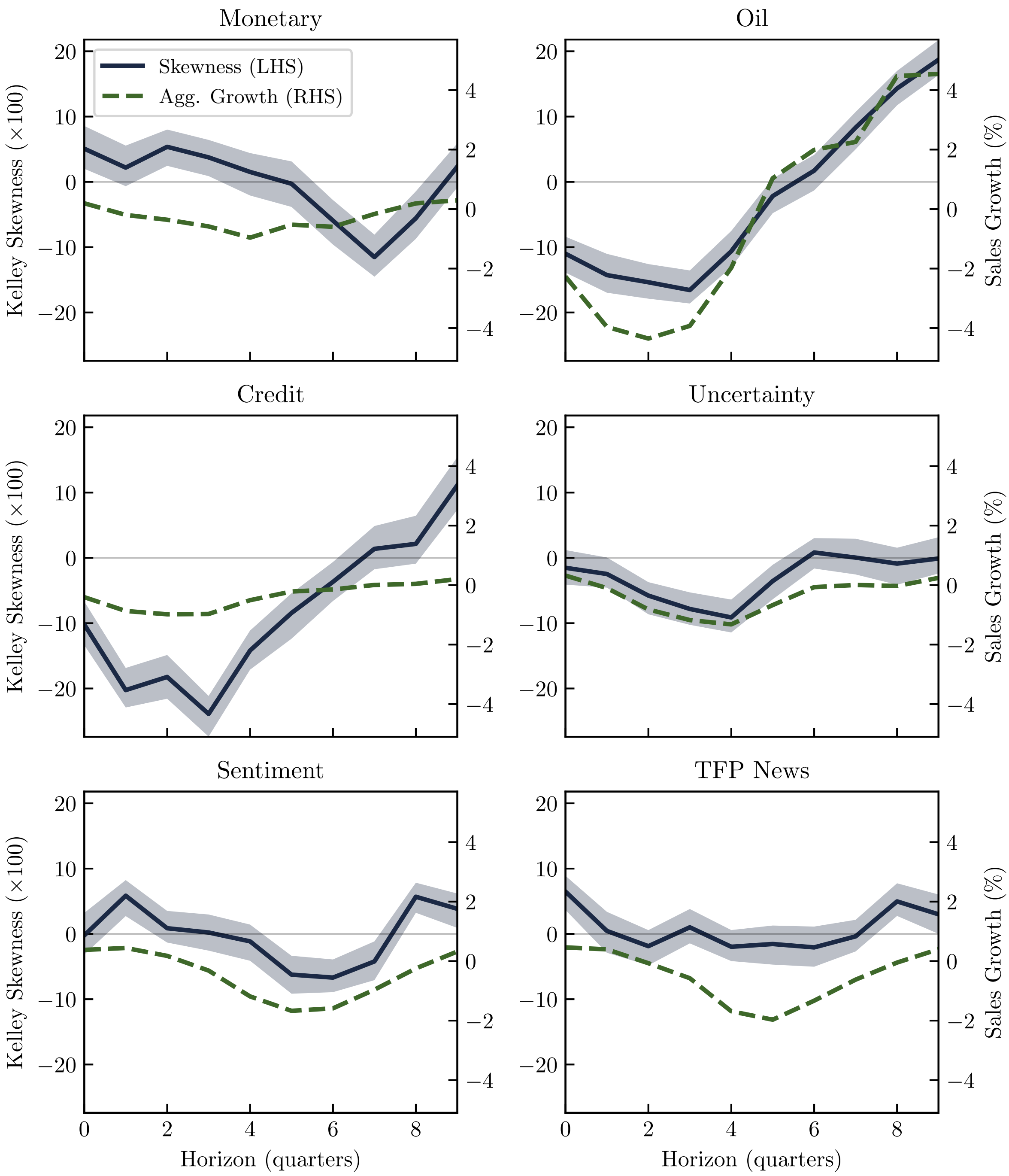
Note: The number of streaks can be larger than the number of unique firms. The average number of time series observations is measured for impact effect regressions and rounded to the nearest integer. The adjusted R^2 values are averaged across horizons and firms. The share of significant IRFs is the relative frequency of statistically significant IRFs for the peak of the impulse response estimates, measured using 90% confidence intervals based on Newey–West standard errors. Quantiles across firm-level IRFs are averaged across horizons. IRFs for the credit shock are only estimated for firms existing during the Great Financial Crisis.

shock-specific controls used in the main local projections; the credit specification also includes contemporaneous GDP growth as in Gilchrist and Zakrajšek (2012).

Table A.3 summarizes the firm-level regressions. The exercise uses more than 4,000 firms for most shocks, except for credit shocks where the sample is restricted to firms observed around the Great Financial Crisis. The estimated responses are widely dispersed, which is exactly the margin the theory emphasizes.

We compute aggregate sales growth as the sales-weighted average of the firm-level IRFs, $\widehat{\beta}_h^{agg} = \sum_i \omega_{\beta,i}\widehat{\beta}_{i,h}$, and cross-sectional skewness as Kelley skewness of the IRF distribution, $\widehat{\beta}_h^{ksk} = ksk(\{\widehat{\beta}_{i,h}\}_i)$. Figure A.5 shows that this bottom-up construction reproduces the main pattern: contractionary shocks lower both aggregate growth and cross-sectional skewness. The comovement is especially clear for oil, credit, and uncertainty shocks; it is weaker but still visible for sentiment and TFP shocks. The monetary response is delayed,

Figure A.5: Comovement of cross-sectional growth and skew after aggregate shocks



Note: The figure shows the response of aggregate sales growth ($\widehat{\beta}_h^{agg}$; green dashed line) and cross-sectional skewness ($\widehat{\beta}_h^{ksk}$; blue solid line) to different aggregate shocks. The blue shaded areas are 90% confidence bands for the skewness response, based on a bootstrap with 2000 replications. Shock magnitudes are normalized to one standard deviation. The signs of the sentiment and TFP shock are reversed to be contractionary.

with skewness reaching its trough after sales growth has already begun to recover.[24]

Figure A.6 confirms the size gradient in the bottom-up exercise. The skewness response among large firms is negative and economically large across shocks, while the response among smaller firms is muted. This is the same pattern as in the top-down skewness indexes in the main text, and it supports the interpretation that aggregate shocks generate skewness through heterogeneous firm responses rather than through one particular aggregate time-series construction.

D.5 Cross-Sectional Skewness v. Aggregate Shocks

D.5.1 Shock Series for IRFs

Table A.4 reports the data sources and transformations used in the local projections. We use published shock series or authors' replication files whenever possible. The main identification choices are: monetary shocks from Bu et al. (2021), which cover conventional and unconventional policy; oil supply shocks from Baumeister and Hamilton (2019); credit shocks from a replication of the Gilchrist and Zakrajšek (2012) excess-bond-premium VAR; financial uncertainty shocks from the maxG solution of Ludvigson et al. (2021); sentiment shocks from the Lagerborg et al. (2023) proxy SVAR; and TFP news shocks from Ben Zeev and Khan (2015). The signs of sentiment and TFP news shocks are reversed in the figures so that all reported shocks are contractionary.

D.5.2 Local projection specifications and robustness

The baseline local projections use the streak sample described in Appendix C. We estimate horizons $h = 0, \ldots, 10$ with two lags of controls, as summarized in Table 2. The main skewness responses are robust to using four lags, adding lagged aggregate sales growth, or computing skewness from the cleaned sample instead of the streak sample.

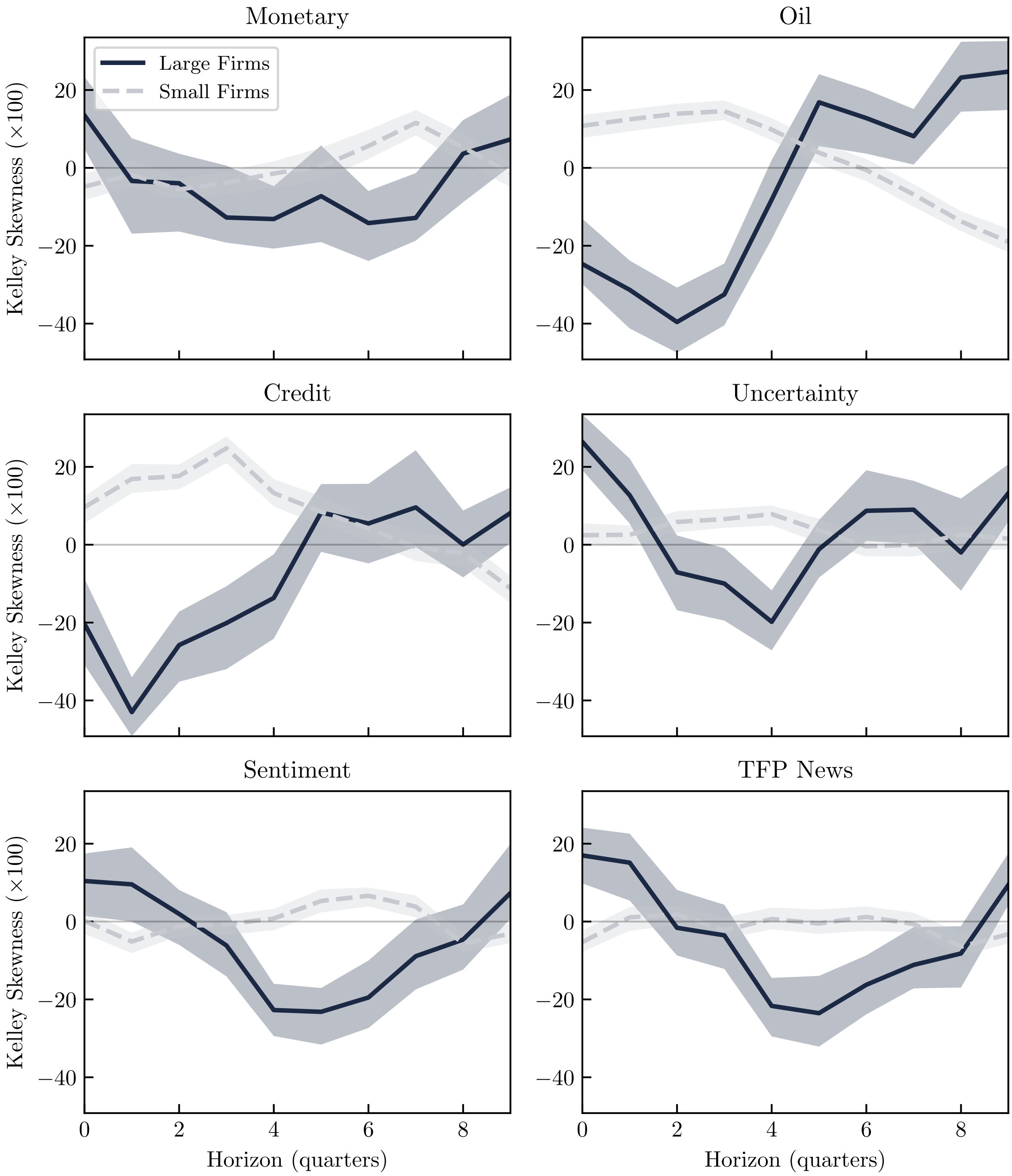
D.6 Results from Compustat Global

We present some complementary analyses to establish robustness of our results. For questions regarding sample construction or shock sequences please consult the supplementary online appendix. The online appendix also includes a bottom-up estimation of impulse-response functions, a growth decomposition documenting skewed aggregate (as opposed to idiosyncratic) components and simulation evidence.

In order to ensure that our main stylized facts are not only present in the US, but more broadly, we present the stylized facts for the Compustat Global Sample (which *excludes*

[24] The bootstrap resamples firm-level IRFs and treats the estimates as fixed, so the bands understate full estimation uncertainty. A full parametric bootstrap would require millions of firm-level regressions.

Figure A.6: Large vs small firms: Cross-sectional skewness responses



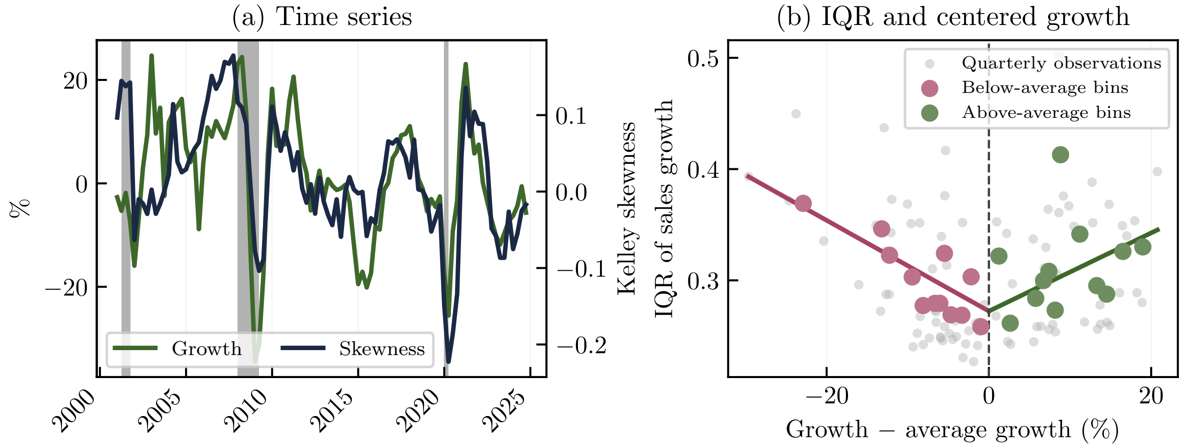
Note: The figure shows responses of cross-sectional skewness ($\widehat{\beta}_h^{ksk}$; blue solid line for large firms and light blue dashed line for small firms) to different aggregate shocks. Large firms are the top 10% of the sales distribution, based on average real sales. Small firms are all other firms. Shaded areas are 90% bootstrap confidence bands. Shock magnitudes are normalized to one standard deviation. The signs of the sentiment and TFP shock are reversed to be contractionary.

Table A.4: Data for local projections

Variable	Transformation from raw data	Data source
Real GDP	Log level	FRED (`GDPC1`)
GDP Deflator	Log level	FRED (`GDPDEF`)
Real oil price	Quarterly average of monthly data, deflated	FRED (`WPU0561`, `GDPDEF`)
GDP per capita	Real GDP (‘rgdp’) per population (‘civpop’)	Ramey (2016) TFP data
Labor productivity	Real GDP (‘rgdp’) per hours worked (‘tothours’)	Ramey (2016) TFP data
Shadow rate	Quarterly average of monthly data	Atlanta Fed*
Stock prices	Shiller stock prices divided by GDP deflator	Ramey (2016), FRED
Stock prices per capita	Stock prices per population (‘civpop’)	Ramey (2016)
VXO	Quarterly average of daily data	FRED (`VXOCLS`)
Uncertainty Index	Log level of Jurado et al. (2015) index	Lagerborg et al. (2023)
Consumer Expectations	Log level	Lagerborg et al. (2023)
Monetary shock	Quarterly sum of monthly data	Fed Board**
Oil shock	Quarterly average of monthly data	Baumeister***
Credit shock	Quarterly average of monthly data	Favara et al. (2016)†
Uncertainty shock	Quarterly average of monthly ‘maxG’ shock	Ludvigson et al. (2021)
Sentiment shock	Quarterly average of monthly data	Lagerborg et al. (2023)‡
TFP News Shock	Level of Ben Zeev and Khan (2015) shock	Ramey (2016) TFP data
Cross-sectional skewness	Own construction based on streak sample	–
Sales growth	Own construction based on streak sample	–

Note:
(*) Shadow rate: `https://www.atlantafed.org/cqer/research/wu-xia-shadow-federal-funds-rate`.
(**) Bu et al. (2021) monetary shocks: `https://www.federalreserve.gov/econres/feds/a-unified-measure-of-fed-monetary-policy-shocks.htm`.
(***) Baumeister and Hamilton (2019) shocks: `https://sites.google.com/site/cjsbaumeister/datasets?authuser=0`.
(†) Credit shock from eight-variable VAR of Gilchrist and Zakrajšek (2012), 1973Q1–2019Q4.
(‡) Sentiment shock from Lagerborg et al. (2023) proxy SVAR, 1965:1–2018:11. Instrument is number of fatalities ($\leq$7), excluding 2017 Las Vegas shooting.

Figure A.7: Growth, Skewness, and Dispersion in Compustat Global

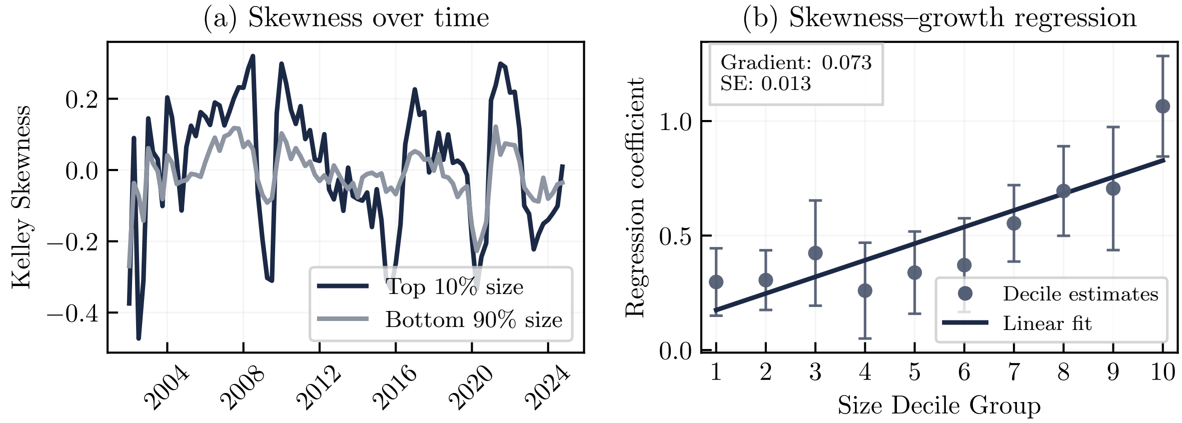


Note: The sample contains non-U.S. firms from Compustat Global over 2000Q1–2024Q4. Sales are converted to U.S. dollars using quarterly average exchange rates and deflated by the U.S. GDP deflator; growth is year-over-year log real sales growth. Panel (a) plots aggregate real sales growth and Kelley skewness. Panel (b) plots the interquartile range of firm sales growth against aggregate growth relative to its sample mean. Gray dots are quarterly observations; colored dots are binned averages below and above average growth. Gray bars in Panel (a) denote NBER recessions.

US companies), between 2000Q1 and 2024Q4. We conduct a similar cleaning procedure as described in Appendix C in order to stay consistent between the two datasets.

Figure A.7 reproduces the main moment patterns in the Compustat Global sample. Panel (a) shows that Kelley skewness of year-over-year sales growth moves closely with aggregate real sales growth. Panel (b) shows the same V-shaped relationship between dispersion and aggregate growth as in the U.S. sample: the interquartile range is lowest near average growth and rises in both contractions and expansions. Figure A.8 then examines the size gradient. Panel (a) shows that skewness fluctuates more strongly among large firms, while Panel (b) shows that the skewness–growth coefficient generally rises across size deciles. The main stylized facts therefore extend to a disjoint sample of non-U.S. firms.

Figure A.8: Size-dependent skewness



Note: Size groups are defined based on average real sales over previous three years. The standard deviation of Kelley skewness for large firms is about 0.23 — more than twice the corresponding value of 0.11 for small firms.

E Simulation

We show that the stylized environment of our formal results carries over into a 'messier' environment. Applying the implied recipe of our theorems, we generate time-series of (1) countercyclical variance and (2) procyclical skewness from a single series of aggregate impulses. The goal is to generate aggregate series that qualitatively look like Figures 2 and 3(a). We use Algorithm 1 which follows the logic of Section 3. The simulation results are shown in Figure 9. They qualitatively agree with our stylized facts. Parameter choices for the simulation algorithm are given by Table A.5.

Table A.5: Simulation Parameters and Functional Definitions

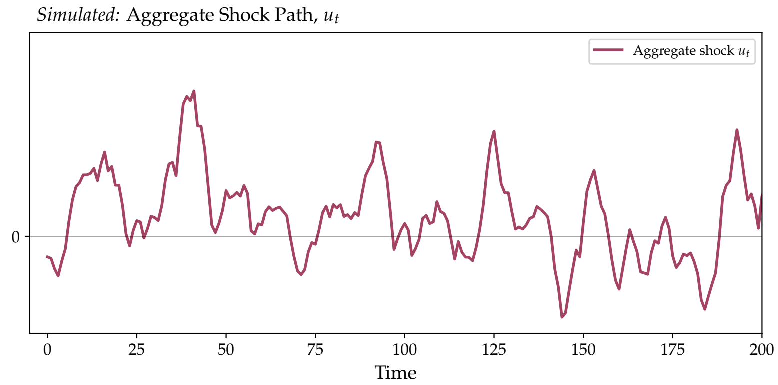
Symbol / Function	**Description**	**Value / Definition**
n	Time-series sample size	500
burn	Burn-in observations	5000
I	Cross-sectional sample size per t	50,000
$\phi_1,\ \phi_2$	AR(2) coefficients	$1.4,\ -0.5$
p	Probability of Pareto innovation	0.04
b	Pareto shape parameter	1.4
s_P	Pareto scale parameter	0.6
$\mu_N,\ \sigma_N$	Mean, std. dev. of normal innovation	$0.1,\ 0.8$
amp	Amplitude of concave map	5
$z_1,\ z_2$	Curvature parameters for $Q_1^\star,\ Q_2^\star$	$6,\ 12$
a	Skew-normal shape parameter	10
loc	Skew-normal location parameter	-0.04
s_S	Skew-normal scale parameter	0.1
$Q_h^\star(x; m, M)$	Concave decreasing map ($h = 1, 2$); m, M are min/max of sample	$\frac{\text{amp}}{z_h}\left[1 - \left(\frac{x-m}{M-m} + 0.5\right)^{z_h}\right]$

Note: $Q_h^\star$ acts elementwise on vectors $\boldsymbol{x}_t$ within the algorithm (vectorized over i).

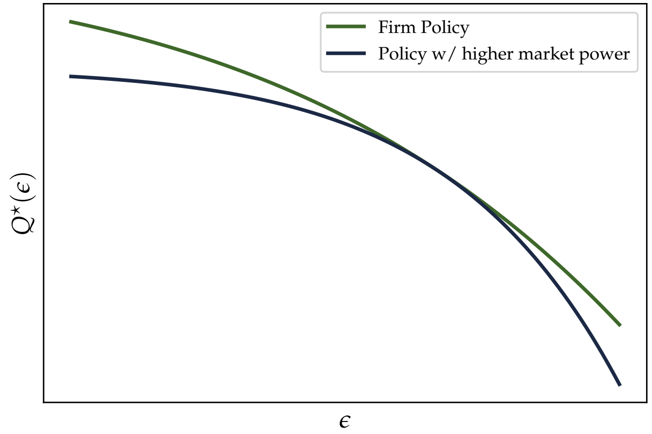
Algorithm 1: DGP for y_t and Cross-Sectional Statistics (vectorized over i)

Aggregate shocks, y_t: AR(2) with normal-Pareto mixture innovations
for $t \leftarrow 1$ **to** n+burn **do**
    Draw $u_t \sim \text{Unif}(0,1)$
    if $u_t < p$ **then** $\varepsilon_t \sim \text{Pareto}(b, \text{scale}{=}s_P)$
    else $\varepsilon_t \sim \mathcal{N}(\mu_N, \sigma_N^2)$
    $y_t \leftarrow \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$
Discard first burn and relabel $t{=}1{:}n$
Firm-level exposures (vectorized)
for $t \leftarrow 1$ **to** n **do**
    Draw $\boldsymbol{x}_t \in \mathbb{R}^I \sim \mathcal{N}\big((y_t/8)\mathbf{1},\ |y_t|^2\mathbf{I}\big)$
$m \leftarrow \min_{t,i} x_{t,i}, \quad M \leftarrow \max_{t,i} x_{t,i}$
Log-output through concave transforms $Q_1^\star, Q_2^\star$
for $t \leftarrow 1$ **to** n **do**
    $\boldsymbol{\ell}_t \leftarrow Q_1^\star(\boldsymbol{x}_t; m, M), \quad \boldsymbol{\ell}'_t \leftarrow Q_2^\star(\boldsymbol{x}_t; m, M)$
Growth rates through quarterly differencing
for $t \leftarrow 5$ **to** n **do**
    $\boldsymbol{g}_t \leftarrow \Delta_4 \boldsymbol{\ell}_t = \boldsymbol{\ell}_t - \boldsymbol{\ell}_{t-4}, \quad \boldsymbol{g}'_t \leftarrow \Delta_4 \boldsymbol{\ell}'_t = \boldsymbol{\ell}'_t - \boldsymbol{\ell}'_{t-4}$
Idiosyncratically skewed trend noise (constant across t per unit)
Draw $\boldsymbol{\eta} \in \mathbb{R}^I \sim \text{SkewNormal}(a, \text{loc}, s_S)$
for $t \leftarrow 5$ **to** n **do**
    $\boldsymbol{g}_t \leftarrow \boldsymbol{g}_t + \boldsymbol{\eta}, \quad \boldsymbol{g}'_t \leftarrow \boldsymbol{g}'_t + \boldsymbol{\eta}$
Cross-sectional stats
for $t \leftarrow 5$ **to** n **do**
    $\bar{g}_t \leftarrow mean(\boldsymbol{g}_t), \quad \bar{g}'_t \leftarrow mean(\boldsymbol{g}'_t)$
    $\text{KellySkew}_t \leftarrow skew(\boldsymbol{g}_t), \quad \text{KellySkew}'_t \leftarrow skew(\boldsymbol{g}'_t)$
Volatility and volatility growth
for $t \leftarrow 1$ **to** n **do**
    $v_t \leftarrow y_t^2 / \max_{s \le n} y_s^2$
for $t \leftarrow 5$ **to** n **do**
    $\Delta_4 v_t \leftarrow v_t - v_{t-4}$

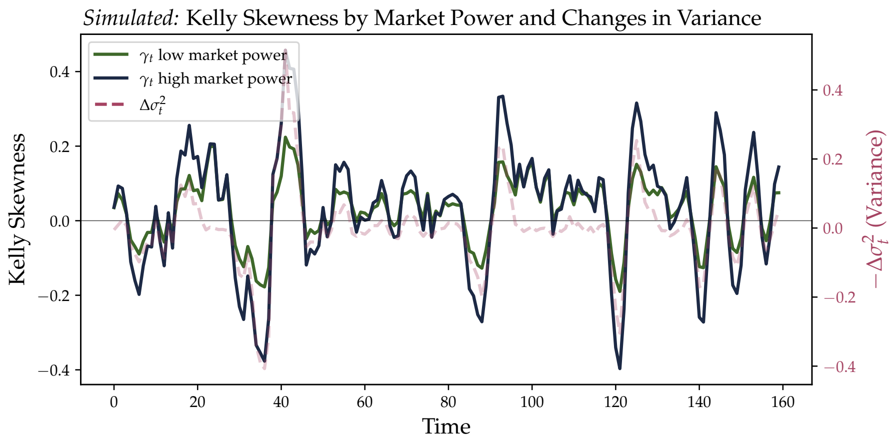
Figure 9: Simulation



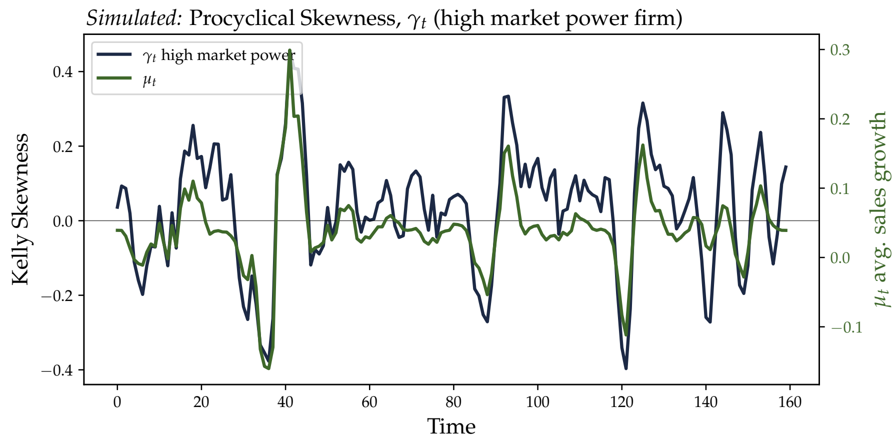
(a) Simulated time-series of a single aggregate factor.



(b) Two concave firm policies.



(c) Co-movement of cross-sectional skewness and changes in variance.



(a) Comovement of growth and skewness.

Note: Simulation procedure of procyclical skewness in pictures. Panel (a) shows an AR(2) process with jumps and slowly decaying IRF as aggregate shock. The aggregate shock scales the mean and the standard deviation of the cross sectional idiosyncratic input shocks, ϵ_i, t (not shown). Panel (b) displays the policy functions for log-output of two firms with different market power. Policy functions map shocks $\epsilon_{i,t}$ to $\widehat{q}$. Growth rates are subject to an additive idiosyncratic growth trend component (secular trend not shown, see algorithm). Panel (c) shows the simulated growth skewness across high- versus low-market power firms, and correlates those with changes in cross-sectional variance of input shocks. Panel (d) shows co-movement of mean and skewness.